Competency Based Learning Material

Sector: **AGRICULTURE, FORESTRY AND FISHERY**	
Qualification : **ORGANIC AGRICULTURE PRODUCTION NC II**	
Unit of Competency: **PRODUCE ORGANIC VEGETABLES**	
Module Title **PRODUCING ORGANIC VEGETABLES**	
MDM SAGAY COLLEGE, INC.	

ORGANIC AGRICULTURE PRODUCTION NC II

The **ORGANIC AGRICULTURE PRODUCTION NC II** Qualification consists of competencies that a person must achieve to produce organic farm products such as chicken and vegetables including producing of organic supplements such as fertilizer, concoctions and extracts. It has two (2) elective competencies which are on raising organic hogs and raising organic small ruminants.

This Qualification is packaged from the competency map of the Agri-Fishery Sector.

The units of competency comprising this qualification includes the following:

CODE	BASIC COMPETENCIES
500311105	Participate in workplace communication
500311106	Work in a team environment
500311107	Practice career professionalism
500311108	Practice occupational health and safety procedures

CODE	COMMON COMPETENCIES
AFF 321201	Apply safety measures in farm operations
AFF 321202	Use farm tools and equipment
AFF 321203	Perform estimation and calculations
TRS311201	Develop and update industry knowledge
AGR321205	Perform record keeping

CODE	CORE COMPETENCIES
AGR612301	Raise organic chicken
AGR611306	Produce organic vegetables
AGR611301	Produce organic fertilizer
AGR611302	Produce organic concoctions and extract
CODE	**ELECTIVE COMPETENCIES**
AGR612302	Raise organic hogs
AGR612303	Raise organic small ruminants

HOW TO USE THIS MODULE

Welcome to the Module on Producing Organic Vegetables. This module contains training materials and activities for you to complete.

The unit of competency on *Produce Organic Vegetables* contains knowledge, skills and attitudes required for an Organic Agriculture NC II course.

You are required to go through a series of learning activities in order to complete each of the learning outcomes of the module. In each learning outcome there are **Information Sheets, Operation Sheets**, **Job Sheets** and **Task Sheets**. Follow these activities on your own and answer the Self-Check at the end of each learning activity.

If you have questions, don't hesitate to ask your trainer for assistance.

Recognition of Prior Learning (RPL)

You may already have some of the knowledge and skills covered in this module because you have:

- been working for some time
- already have completed training in this area.

If you can demonstrate to your trainer/instructor that you are competent in a particular skill or skills, talk to him/her about having them formally recognized so you don't have to do the same training again. If you have a qualification or Certificate of Competency from previous trainings show it to your trainer. If the skills you acquired are still current and relevant to this module, they may become part of the evidence you can present for RPL. If you are not sure about the currency of your skills, discuss it with your trainer.

After completing this module ask your trainer to assess your competency. Result of your assessment will be recorded in your competency profile. All the learning activities are designed for you to complete at your own pace.

Inside this module you will find the activities for you to complete followed by relevant information sheets for each learning outcome. Each learning outcome may have more than one learning activity.

MODULE CONTENT

Program/Course : Organic Agriculture Production NC II
Unit of Competency : Produce Organic Vegetables
Module : Producing Organic Vegetables

INTRODUCTION:

This module contains information and suggested learning activities on producing organic vegetable. It includes activities and materials on Organic Agriculture Production NC II

Completion of this module will help you better understand the succeeding module on the raising organic chicken.

This module consists of 4 learning outcomes. Each learning outcome contains learning activities supported by each instruction sheets. Before you perform the instructions, read the information sheets and answer the self-check and activities provided to ascertain to yourself and your trainer that you have acquired the knowledge necessary to perform the skill portion of the particular learning outcome.

Upon completion of this module, report to your trainer for assessment to check your achievement of knowledge and skills requirement of this module. If you pass the assessment, you will be given a certificate of completion.

SUMMARY OF LEARNING OUTCOMES:

Upon completion of the module you should be able to:

LO1 Establish Nursery
LO2 Plant Seedlings
LO3 Perform Plant Care and Management
LO4 Perform harvest and Post-harvest activities

Learning Outcome #1 Establish Nursery

Assessment Criteria:

Seeds are selected in accordance with the PNS, and NSQCS/BPI.

Seedbeds are prepared in accordance with planting requirements based on Vegetable Production manual (VPM).

Care and maintenance of seedlings are done in accordance with enterprise practice.

Potting media are prepared in accordance with enterprise procedure.

Contents:

Selection of Seeds
Seed bed Preparation
Maintaining Seedling
Prepare growing media

Tools, Materials and Equipment and Facilities

Farm
Different vegetable seeds
Seed bed
Carbonized rice hull
Compost
Animal manure
PPE

References:

https://www.hortmag.com/weekly-tips/how-to-know-if-garden-seed-is-viable

https://www.gardenersworld.com/how-to/grow-plants/how-to-test-seed-viability/

https://www.nrcs.usda.gov/Internet/FSE_PLANTMATERIALS/publications/idpmctn10748.pdf

https://smartlivingidea.com/how-to-prepare-a-seedbed/

http://www.heirloom-organics.com/guide/seedstartingcare.html

https://askinglot.com/how-do-you-maintain-seedlings

https://avrdc.org/prepare-growing-medium-seedling-production/

LEARNING EXPERIENCES/ACTIVITIES

LO # 1 ESTABLISH NURSERY

Learning Activities	Special Instructions
Read the attached Information Sheet 2.1-1 on Selection of Seeds	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 2.1-1 on Selection of Seeds	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 2.1-1 on Selection of Seeds	
Perform Job Sheet 2.1-1 on Selection of Seeds	Observe the demonstration of your trainer and check the procedures while performing the operation.
Evaluate performance using performance checklist on 2.1-1 on Selection of Seeds	
Read the attached Information Sheet 2.1-2 on Seedbed Preparation	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 2.1-2 on Seedbed Preparation	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 2.1-2 on Seedbed Preparation	
Perform Job Sheet 2.1-2 on Seedbed Preparation	Observe the demonstration of your trainer and check the procedures while performing the operation.
Evaluate performance using performance checklist on 2.1-2 on Seedbed Preparation	

Read the attached Information Sheet 2.1-3 on Maintaining Seedling	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 2.1-3 on Maintaining Seedling	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 2.1-3 on Maintaining Seedling	
Perform Job Sheet 2.1-3 on Maintaining Seedling	Observe the demonstration of your trainer and check the procedures while performing the operation.
Evaluate performance using performance checklist 2.1-3 on Maintaining Seedling	
Read the attached Information Sheet 2.1-4 on Prepare growing media	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 2.1-4 on Prepare growing media	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 2.1-4 on Prepare growing media	
Perform Job Sheet 2.1-4 on Prepare growing media	Observe the demonstration of your trainer and check the procedures while performing the operation.
Evaluate performance using performance checklist 2.1-4 on Prepare growing media	

After doing this activity, you are now ready to proceed to the next module on Plant seedlings.

INFORMATION SHEET 2.1-1

Seeds Selection

Learning Objective: After reading this information sheet, you should be able to select viable seeds.

Introduction:

Seed is a basic input in agriculture. Strictly speaking seed is an embryo, a living organism embedded in the supporting or the food storage tissue. In seed, the importance is given to the biological existence whereas; in grain the importance is given to the supporting tissue the economic produce.

As per Seed Act (1966) seed includes

- Seed of food crops including edible oil seeds and seeds of fruits & vegetables.
- Cotton seeds
- Seeds of cattle fodder
- Jute seeds
- Seedlings, tubers, bulbs, rhizomes, roots, cuttings, all types of grafts and other vegetative propagated material for food crops (or) cattle fodder

Physical Characteristics of a Good Seeds

1. No insect damage
2. No physical damage (cracks, bruises, etc.)
3. No deformities
4. No disease damage

Importance of quality seed

- Ensures genetic and physical purity of the crops
- Gives desired plant population
- Capacity to withstand the adverse conditions

- Seedlings produced will be more vigorous, fast growing and can resist pest and disease incidence to certain extent
- Ensures uniform growth and maturity
- Development of root system will be more efficient that aids absorption of nutrients efficiently and result in higher yield.
- It will respond well to added fertilizer and other inputs.
- Good quality seeds of improved varieties ensure higher yield at least 10 – 12 %

Major seed quality characters

Seed quality is the possession of seed with required genetic and physical purity that is accompanied with physiological soundness and health status. The major seed quality characters are summarized as below.

1. **Physical Quality**: It is the cleanliness of seed from other seeds, debris, inert matter, diseased seed and insect damaged seed. The seed with physical quality should have uniform size, weight, and color and should be free from stones, debris, and dust, leafs, twigs, stems, flowers, fruit well without other crop seeds and inert material. It also should be devoid of shriveled, diseased mottled, molded, discolored, damaged and empty seeds. The seed should be easily identifiable as a species of specific category of specific species. Lack of this quality character will indirectly influence the field establishment and planting value of seed. This quality character could be obtained with seed lots by proper cleaning and grading of seed (processing) after collection and before sowing / storage.
2. **Genetic purity:** It is the true to type nature of the seed. i.e., the seedling / plant / tree from the seed should resemble its mother in all aspects. This quality character is important for achieving the desired goal of raising the crop either yield or for resistance or for desired quality factors.
3. **Physiological Quality:** It is the actual expression of seed in further generation / multiplication. Physiological quality characters of seed

comprises of seed germination and seed vigor. The liveliness of a seed is known as viability. The extent of liveliness for production of good seedling or the ability of seed for production of seedling with normal root and shoot under favorable condition is known as germinability. Seed vigor is the energy or stamina of the seed in producing elite seedling. It is the sum total of all seed attributes that enables its regeneration of under any given conditions. Seed vigor determines the level of performance of seed or seed lot during germination and seedling emergence. Seed which perform well at sowing are termed as quality seed and based on the degree of performance in production of elite seedling it is classified as high, medium and low vigor seed. The difference in seed vigor is the differential manifestation of the deteriorative process occurring in the seed before the ultimate loss of ability to germinate. Difference in seed vigor will be expressed in rate of emergence, uniformity of emergence and loss of seed germination. Hence it is understood that all viable seeds need not be germinable but all germinable seed will be viable. Similarly, all vigorous seeds will be germinable but all germinable seed need not be vigorous. Physiological quality of seed could be achieved through proper selection of seed (matured seed) used for sowing and by caring for quality characters during extraction, drying and storage. Seed with good vigor is preferable for raising a good plantation as the fruits, the economic come out are to be realized after several years. Hence selection of seed based on seed vigor is important for raising perfect finalize plantation.

4. **Seed Health** - Health status of seed is nothing but the absence of insect infestation and fungal infection, in or on the seed. Seed should not be infected with fungi or infested with insect pests as these will reduce the physiological quality of the seed and also the physical quality of the seed in long term storage. The health status of seed also includes the deterioration status of seed which also expressed through low vigor status of seed. The health status of seed influences the seed quality characters directly and

warrants their soundness in seed for the production of elite seedlings at nursery / field.

Hence the quality seed should have

- High genetic purity
- High pure seed percentage (physical purity)
- High germinability
- High vigor
- Higher field establishment
- Free from pest and disease
- Good shape, size, color etc., according to the specification of variety
- High longevity / shelf life.
- Optimum moisture content for storage
- High market value

Characteristics of good quality seed

- Higher genetically purity:
  - Breeder /Nucleus - 100%
  - Foundation seed - 99.5%
  - Certified seed - 99.0%
- Higher physical purity for certification
  - Maize , Bhendi - 99%
  - All crops (most) - 98%
  - Carrot - 95%
  - Sesame, soybean & jute - 97 %
  - Ground nut - 96 %
- Possession of good shape, size, color, etc., according to specifications of variety
- Higher physical soundness and weight
- Higher germination (90 to 35 % depending on the crop)

- Higher physiological vigor and stamina
- Higher storage capacity
- Free from other crop seeds (Expressed in number /kg) - Other crop seeds are the plants of cultivated crops found in the seed field and whose seed are so similar to crop seed that is difficult to separate them economically by mechanical means. Eg. Mixtures of Wheat, oats seeds in barley.
- It should be free from objectionable weed seeds -These are plants of weed species which are harmful in one or more of the following ways.
  - The size and shape of their seeds are so similar to that of the crop seed that is difficult to remove their seed economically by mechanical means.
  - Their growth habit is detrimental to the growing seed crop due to competing effect.
  - Their plant parts are poisonous or injurious to human and animal beings
  - They serve as alternate hosts for crop pests and diseases.
- It should be free from designated diseases - It refers to the diseases specified for the certification of seeds and for which certification standards are to be met with. These diseases would cause contamination, when they are present in the seed field or with in the specified isolation distance (eg. loose smut of wheat). For this the certification distance has been prescribed as 180 meters.

Crop	Designated disease
Wheat	Loose smut
Sorghum	Grain smut, Kernel smut
Mustard	Alternaria blight

Pearl millet	Grain smut, Green ear Ergot
Sesame	Leaf spot
Brinjal	Little leaf
Chilies	Anthracnose leaf blight, Leaf blight
Cucurbits	Mosaic
Cowpea	Anthracnose
Bhendi	Yellow vein mosaic
Potato	Brown rot , Root knot nematode
Tomato	Early blight, Leaf spot

- It should have optimum moisture content for storage - Long term storage - 6 - 8 % , Short term storage - 10 - 13%
- It should have high market value

Seed selection is very important step in cultivation, quality of seeds should always be good so that we may get better production of crops.

SELF CHECK 2.1-1

SELECTION OF SEEDS

1. _________What is the cleanliness of seed from other seeds, debris, inert matter, diseased seed and insect damaged seed.

2. _________What is the actual expression of seed in further generation / multiplication.

3. ________What is a basic input in agriculture.

4. __________What is the true to type nature of the seed. i.e., the seedling / plant / tree from the seed should resemble its mother in all aspects.

5. ___________What is the possession of seed with required genetic and physical purity that is accompanied with physiological soundness and health status.

ANSWER KEY 2.1-1

Seeds Selection

1. Physical Quality
2. Physiological Quality
3. Seed
4. Genetic Purity
5. Seed Quality

JOB SHEET 1.1-1

Seeds Selection

Title: Selecting viable seeds
Performance Objective: At the end of this module the student should be able to identify and select viable seeds.
Supplies/Materials: PPE, different kind of vegetable seeds.
Steps/Procedures: Wear the PPE Prepare the different kinds of vegetable seeds Observe and Examine its physical characteristics Identify which seeds are viable Do housekeeping
Assessment Method: Demonstration

PERFORMANCE CHECKLIST 2.1-1

Seeds Selection

Performance Criteria	YES	NO
Did the trainee ..		
1. Wear the PPE?		
2. Prepare the different kinds of seeds?		
3. Observe and examine its physical characteristics?		
4. Identify which seeds are viable?		
5. Do housekeeping?		

INFORMATION SHEET 2.1-2

Seedbed Preparation

Learning Objective: After reading this information sheet, you should be able to prepare seedbed.

Introduction:

SEEDBED PREPARATION
Seedbed – or seedling bed is the local soil environment in which seeds are planted.

Seedbed preparation is an important step that can optimize seed germination and survival rate. Final seedbeds can be prepared by disk, harrow, or chisel plowing. The means of preparation will vary with the species selected and the availability of equipment. Unlike standard agricultural settings, reconstructed soils produce greater yields when disked or chisel plowed (Powell, 1988). These treatments are presumed to improve water infiltration rates and reduce the bulk density of the rooting media following soil reconstruction. Similarly, single disking was demonstrated to produce greater yields than multiple diskings (Powell, 1988).

Proper seedbed preparation is vital to successful forage stand establishment. The soil and the planting technique must assure that good soil-seed contact is achieved. If plowed, then the soil should be disked and compacted with a corregated roller before seeding occurs. If no-till planting is practiced, the tilling operation and compaction are applied in a limited area during the planting operation. Precipitation or irrigation of plowed, disked, and harrowed soils may negate the need for compaction prior to planting. In either case, the seedbed must be firm and compact to assure optimum seed-soil contact. A rule-of-thumb for soil surface firmness is to have the soil sufficiently firm that when a person stands on it, the indentation caused by the weight is about 1 cm (approximately 3/8 in.) deep.

Land/seedbed preparation

The method of seedbed preparation differs for conservation (reduced or zero-till) and conventional-till systems. But, for both, the seedbed should be free of weeds and precisely leveled at the time of sowing. For conventional-till dry drill seeding

(CT-dry-DSR), the soil should be well pulverized to maintain good soil moisture for drilling and good soil-to-seed contact. In sandy or silt loam, an excellent seedbed can be prepared with reduced or minimum tillage, thereby conserving soil, and reducing cost. In zero-till dry drill seeding (ZT-dry-DSR), it is important to first knock down the existing vegetation (annual and perennial weeds) with a burn down herbicide such as paraquat (0.5 kg ai ha^{-1}) or glyphosate (1.0 kg ai ha^{-1}).

Tillage and Cultivation

Tillage, for seedbed preparation, and cultivation, for weed control, can affect established plants and reproductive structures. Tillage practices can range from nearly complete soil inversion, with mold-board plowing, to minimal soil disruption, with the use of zero-tillage (direct drilling) techniques. In addition to determining characteristics of soil disturbance and residue incorporation, tillage practices can have important effects on weed density and species composition (Buhler, 1995; Froud-Williams, 1988). Factors determining tillage effects on weeds include (i) depth of seed burial, (ii) seed survival at different soil depths, (iii) seed dormancy responses to burial, (iv) seedling ability to emerge from different burial depths, and (v) the quantity of new seeds added to the soil seedbank (Mohler, 1993). Because weed species may differ in these factors, responses to tillage are often species specific. For example, in a study comparing moldboard plow, chisel plow, and no-tillage systems for soybean production, Buhler and Oplinger (1990) observed that lambsquarters densities were not greatly influenced by tillage systems, whereas redroot pigweed densities were generally highest in the chisel plow system. Giant foxtail *(Setaria faberi* Herrm.), an annual grass species, was most abundant in the no-tillage system and least abundant in the moldboard plow system. In contrast, velvetleaf, an annual broadleaf species, was most abundant in the moldboard plow system and least abundant in the no-tillage system. Similarly, in corn production systems, Buhler (1992) observed that density responses to tillage differed among weed species. Derksen *et al.* (1993) have suggested that changes in weed communities are influenced more by location and year than by tillage systems, but it appears that tillage is potentially useful for reducing weed density if choice among tillage practices is based on knowledge of the full spectrum of ecological sensitivities of different weed species.
Because seeds of many weed species require exposure to light to germinate, attention has been directed recently toward the possibility of preparing crop seedbeds at night or during the day using tillage equipment covered with light-excluding hoods. In field trials Ascard (1994) observed that, compared to daylight tillage, both of the aforementioned dark-tillage techniques reduced weed density.

In some farming areas, germination of common weed species may occur in predictable flushes that are driven by accumulated heat units and rainfall (e.g., Harvey and Forcella, 1993). Because weed seedlings are extremely vulnerable to tillage operations, Forcella *et al.* (1993) conducted experiments in the north central United States to determine whether synchronic-ity in weed emergence could be exploited for management purposes. Delaying final tillage operations—the so-called stale seedbed strategy—allowed the investigators to kill a very high percentage of weed seedlings before planting corn and soybean and, consequently, to reduce weed competition against the crops.
Following planting, cultivation can be used before and after crop emergence to reduce weed densities between and within crop rows (Terpstra and Kouwenhoven, 1981; Buhler *et al,* 1992; Rasmussen, 1992; Rydberg, 1993; Mulder and Doll, 1994; Rasmussen and Svenningsen, 1995; Vangessel *et al,* 1995). Increased interest in alternatives to herbicides has resulted in a number of new cultivation implements, some of which are capable of working under high residue conditions and therefore are compatible with soil conservation objectives (Eadie *et al,* 1992). Interest has also increased in machinery for flame weeding, which kills weed seedlings through cell rupture rather than incineration (Daar, 1987). Flame weeding can be used to destroy weeds emerging before crop emergence (Ascard, 1995a,b); post-emergence flaming, long practiced in cotton (*Gossypium hirsutum* L.), is also possible in certain other crops such as onion (*Allium cepa* L.) and corn (Daar, 1987).
Ploughing is the first operation in seedbed preparation on most farms and is likely to remain so for some time yet. However, many farmers are now using rotary cultivators, heavy cultivators with fixed or spring tines, and mechanically-driven digging or pulverizing machines, as alternatives to the plough. Good ploughing is still the best method of burying weeds and the remains of previous crops. It can also set up the soil so that good frost penetration is possible. Fast ploughing produces a more broken furrow slice than slow steady work. The mounted or semi-mounted plough has replaced the trailed type on most farms because of ease of handling. General-purpose moldboards are commonly used. The shorter digger types (concave moldboards) break the furrow slices better and are often used on the lighter soils. Deep digger ploughs are used where deep ploughing is required, e.g. for roots or potatoes. The one way (reversible) type of plough has become popular for crops such as roots and peas. It has right-hand and left-hand moldboards and so no openings or finishes have to be made when ploughing; the seedbed should therefore be at least slightly more level. Round-and-round ploughing with the ordinary plough has almost the same effect, although this is not a suitable method on all fields.
The proper use of skim and disc coulters and careful setting of the plough for depth, width and pitch can greatly improve the quality of the ploughing. The furrow slice can only be turned over satisfactorily if the depth is less than about

two-thirds the width. The usual widths of ordinary plough bodies vary from 20 to 35 cm. If possible, it is desirable to vary the depth of ploughing from year to year to avoid the formation of a plough pan. Very deep ploughing, which brings up several centimetres of poorly weathered subsoil, should only be undertaken with care: the long-term effects will probably be worthwhile but, for a few years afterwards, the soil may be rather sticky and difficult to work. Buried weed seeds, such as wild oats which have fallen down cracks, may be brought to the surface and may spoil the following crops. 'Chisel ploughing' is a term used to describe the work done by a heavy duty cultivator with special spring or fixed tines; unlike the ordinary plough, it does not move or invert all the soil. Disc ploughs have large saucer-shaped discs instead of shares and moldboards. Compared with the ordinary moldboard plough, they do not cut all the ground or invert the soil so well, but they can work in harder and stickier soil conditions. They are more popular in dry countries. Double moldboard ridging ploughs are sometimes used for potatoes and some root crops in the wetter areas.

SELF CHECK 2.1-2

Seedbed Preparation

1. _______________ What is the first operation in seedbed preparation?

2. _______________What is an important step that can optimize seed germination and survival rate?

3. _______________ What is a term used to describe the work done by a heavy duty cultivator with special spring or fixed tines; unlike the ordinary plough, it does not move or invert all the soil?

4. _______________ What is the local soil environment in which seeds are planted?

5. ______________ What ploughs are sometimes used for potatoes and some root crops in the wetter areas ?

ANSWER KEY 2.1-2

Seedbed Preparation

1. Ploughing
2. Seedbed preparation
3. Chisel ploughing’
4. Seedbed
5. Double moldboard ridging ploughs

JOB SHEET 2.1-2

Seedbed Preparation

Title: Preparing Seedbed
Performance Objective: At the end of this module the student should be able to prepare seedbed.
Supplies/Materials: PPE, harrow, field
Steps/Procedures: Wear the PPE Gather tools and materials Visit the field Clearing the field Perform Ploughing Do housekeeping
Assessment Method: Demonstration

PERFORMANCE CHECKLIST 2.1-2

Seedbed Preparation

Performance Criteria	YES	NO
Did the trainee ..		
1. Wear the PPE?		
2. Gather tools and materials?		
3. Visit the field?		
4. Clearing the field?		
5 Perform ploughing?		
6 Do housekeeping?		

INFORMATION SHEET 2.1-3

Maintaining Seedling

Learning Objective: After reading this information sheet, you should be able to care and maintain seedlings.

Introduction:

Maintaining your seedling

As your seedling emerges from the soil, most grower breath a big sigh of relief. Close your eyes and you can almost see the plant grow and flourish into its full beauty, producing an enviable profusion of flowers or vegetables. As you open your eyes, you will immediately begin a new set of worries, over-nurturing the newborn indoors for a short time while the outdoor weather catches up with your dreams.

Thin Seedlings as needed
Plants in your garden do not like to be crowded. Ditto with your seedlings, who need all the sun and nutrients that they can get you may want to leave a few extras for a while as mortality rate of seedlings can be high.

Give them plenty of light
As soon as the new born seedling begins to emerge, it seeks light. Your new born needs as much and as direct a light source as possible. Placing it by a window with a southern exposure is the first step. But this alone may not prove to be enough for the seedling to grow healthy and strong. First, the sun is not up as long in the spring as it is in the summer. Second, there are many rainy spring days with little or no direct sun. You should also acquire an artificial Grow Light and place the seedlings under it on cloudy days and at night.

Keep the seedlings moist
Provide water to your seedling every couple of days. Do not soak the soil each night. Overly wet soil encourages the development of damping off disease. Let the soil dry out a little on the top, then water thoroughly.

Watering from the bottom is preferred. If you have a seed tray, add water to the bottom of the tray . The soil will absorb it through the bottom holes in your container...your container does have holes in the bottom, doesn't it!?!

Feed the seedlings
The seedling does not need a lot of extra nutrients in it's first few days of life. Your soil starting mix usually comes with a balanced formula of nutrients that the seedlings need. After several days, adding a little liquid fertilizer to the water is helpful, but you do not need to give it full strength.

If the roots begin to come out the bottom of the pot, it is time to plant your seedling outdoors, weather permitting. If it is still too cool, keep the bottom of the tray moist, or put some extra soil in the bottom of the tray. Or, transplant seedlings to a larger pot. Most plants do not like to be root bound.

Guard against Leggy Plants
Seedlings are leggy when their main stem or stalk grows tall and thin and can hardly support the leaf structure. It is caused by insufficient sunlight and a sheltered environment. Indoors, they do not experience the effect of wind, and do not need to develop structure to defend against it. Most seedlings do not even experience a slight breeze. When transplanted outdoors, "leggy" plants can be damaged or broken by the wind.

> **Tip:** Take your hand,or a couple sheets of newspaper and fan the plants a few times a day. You can even lightly brush the tops of the plants, brushing back and forth in varying directions. You may notice the plants seem to slow down for a period. What they are really doing is building a stronger stem or stalk.

Protect Against Damping Off Disease

Those of us who have grown seedling indoors for any number of years know what "Damping Off" disease. This is a white mold that forms in the top of the soil. Damping Off disease flourishes in cold, wet damp weather along with little sunshine. It quickly spreads across the soil and wilts the seedling. Take its habitat away, and the disease cannot survive. Plants on the other hand, love just the opposite conditions. The more you make conditions ideal for your plants, the more likely you will avoid Damping Off Disease and other fungal problems.

If you do experience problems, do not give up hope. Here are some things you can do to minimize or eliminate disease problems:

- First, get the plant in direct sunlight if at all possible.
- Stop watering until the surface is very dry.
- Water only from the bottom.
- Scrape as much of the mold off the soil as possible.
- Stir the top of the soil without disturbing the roots. It will also speed drying.
- Add some soil, although this may or may not produce results.
- Increase room air circulation. You can gently blow air on your plant trays with a small fan.

Avoid sowing your seeds in the basement and leaving them there for a couple of days. While the trays are conveniently out of the way, this is a perfect breeding ground for Damping Off Disease.

What exactly is Damping Off Disease?

Somewhere lurking in the air in your house is the fungus spores of the most dreaded of plant disease for those of us who start plants indoors for transplanting outdoors later in the season.

Damping off Disease is very common plant disease problem. We fear it, because it is fatal to our young seedlings, and is quite harmful to our soaring spring spirits. To lose seedlings so early in the new gardening year is just heart breaking, especially if it is a special seed. It leads to replanting, and gets our young gardening season off to a late start.

If you grow indoor transplants early in the spring, you likely have experienced it at some point. We usually think of Damping Off Disease as an indoor plant problem. But, it also occurs outdoors, too. We are less likely to recognize it outdoors, as the loss of plants in the spring can be attributed to a number of things.

Now for the good news.... Damping Off Disease as a threat to your seedlings can be minimized. We have lots of tips and ideas to help fight off this enemy of the state.

Causes of Disease

Damping Off disease thrives in cool or cold, dark or cloudy, wet or damp conditions. The disease is airborne, and can spread very quickly from one seed tray to another.

The fungal spores take root in your soil and quickly spreads across the seed tray, jumping to other trays with ease. It is fatal to young seedlings, nipping them off at the soil level.

Treatment
As with other plants diseases, prevention is the best means of treatment. Follow the do's and don't's listed below. If Damping Off disease does take hold in your seed trays, act immediately. Remove diseased sections to minimize the spread. If it has affected a significant number of plants, replant in new soil and clean containers. Do not reuse the soil. Either use new containers, or sterilize the ones you were using. We recommend new containers.

Controlling the Disease
Controlling the disease is a matter of removing the environment that Damping Off disease thrives in. Here are the basic **do's** and **don'ts**:

Do's	**Don'ts**
Do buy sterilized seed starting soil.	**Don't** leave your seedling trays in the basement. Basements are perfect breeding grounds.
Do use clean, sterilized containers.	**Don't** overwater plants.
Do use clean, sterilized containers.	**Don't** use fertilizer on your new seedlings.
Do provide plenty of air circulation.	**Don't** use tray covers. While it is a popular practice to use them, they increase the humidity level and encourage disease growth.
Do use a small fan and direct a gentle breeze across the room. The important word here is "gentle"	Did you know? Nitrogen in your fertilizer can promote rapid growth of Damping Off Disease.
Do thin seedlings to increase air circulation.	
Do provide as much sunlight as possible.	
Do let the surface of the soil dry out between watering. Watering from the bottom is preferred.	
Do stir the top of the soil around the seedlings.	

Other Tips and Suggestions:
It is believed that soaking seeds in a small amount of water and a clove of crushed garlic will prevent the disease.

Some people suggest misting the plant with Chamomile tea as a preventative.

Some people suggest fireplace ash on the top of the soil.

Cinnamon also acts as a fungicide.

Sphagnum moss spread thinly on the surface of the soil.

SELF CHECK 2.1-3

Maintaining Seedlings

TRUE OR FALSE

1. ____________________ Don’t use clean, sterilized containers.

2. ____________________ Do thin seedlings to increase air circulation.

3. __________________ Seedlings should be watered once a day.

4. __________________ The seedling does not need a lot of extra nutrients in it's first few days of life.

5. __________________Damping Off disease thrives in cool or cold, dark or cloudy, wet or damp conditions.

ANSWER KEY 2.1-3

Maintaining Seedlings

1. False
2. True
3. True
4. True
5. True

JOB SHEET 2.1-3

Maintaining Seedlings

Title: Maintaining Seedlings
Performance Objective: At the end of this module the student should be able to maintain of seedlings.
Supplies/Materials: PPE**,** Trowel, Sprinkler, Container, water, seedlings
Steps/Procedures: Wear the PPE Gather the tools and materials Check the seedlings, if there's unhealthy one separate it immediately Perform thinning Water the seedlings Do housekeeping
Assessment Method: Demonstration

PERFORMANCE CHECKLIST 2.1-3

Maintaining Seedlings

Performance Criteria	YES	NO
Did the trainee ..		
1. Wear the PPE?		
2. Gather the tools and materials?		
3. Check the seedlings, if there's unhealthy one separate it immediately?		
4. Perform thinning the seedlings?		
5. Water the seedlings?		
6. Do housekeeping?		

INFORMATION SHEET 2.1-4

Prepare Growing Media

Learning Objective: After reading this information sheet, you should be able prepare the different growing media.

Introduction:

Prepare Growing Media

The production of greenhouse crops involves a number of cultural inputs. Among these, perhaps the most important is the type of growing medium used. Due to the relatively shallow depth and limited volume of a container, growing media must be amended to provide the appropriate physical and chemical properties necessary for plant growth.

Field soils are generally unsatisfactory for the production of plants in containers. This is primarily because soils do not provide the aeration, drainage and water holding capacity required. To improve this situation several “soilless” growing media have been developed. The following is a description of some of the most commonly used amendments for the production of greenhouse crops.

Peat and Peat-Like Materials

Peat moss is formed by the accumulation of plant materials in poorly drained areas. The type of plant material and degree of decomposition largely determine its value for use in a growing medium. Although the composition of different peat deposits vary widely, four distinct categories may be identified:

Hypnaceous moss – this type of peat consists of the partially decomposed remains of hyprum, polytrichum and other mosses of the Hypanaceae family. Although it decomposes more rapidly than some other peat types, it is suitable for media use. Many of the peat deposits in the Northern United States are Hypnaceous.
Reed and Sedge – are peats derived from the moderately decomposed remains of rushes, coarse grasses, sedges, reeds and similar plants. These fine textured materials are generally less acid and contain relatively few fibrous particles. The rapid rate of decomposition, fine particle size and insufficient fiber content make reed and sedge peats unsatisfactory for media use.

Humus or Muck – consists of the decomposed debris of finely divided plant materials of unknown origin. Humus often contains large quantities of silt and clay particles, and when mixed with soil does not improve drainage or aeration. Due to its rapid rate of decomposition and particle size, humus is considered to be undesirable for growing media use.
Sphagnum moss – is the dehydrated remains of acid-bog plants from the genus Sphagnum (i.e. Spapillosum). It is light in weight and has the ability to absorb 10 to 20 times its weight in water. This is attributed to the large groups of water holding cells, characteristic of the genus. Sphagnum moss contains specific fungistatic substances which accounts for its ability to inhibit damping-off of seedlings.

Sphagnum moss is perhaps the most desirable form of organic matter for the preparation of growing media. Drainage and aeration are improved in heavier soils while moisture and nutrient retention are increased in lighter soils. Germany, Canada and Ireland are the principle regions of Sphagnum moss production.

Wood Residues

Wood residues constitute a significant source of soilless growing media. These materials are generally bi-products of the lumber industry and are readily available in large quantities. Nitrogen depletion by soil microorganisms, during the decomposition process, is one of the primary problems associated with these materials. However, supplemental applications of N to the growing media can make most wood residues valuable amendments.

Leaf Mold – maple, oak, and sycamore are among the principle leaf types suitable for the preparation of leaf mold. Layers of leaves and soil are composted together with small amounts of nitrogenous compounds for approximately 12 to 18 months. The use of leaf mold can effectively improve the aeration, drainage and water holding properties of a growing media. Although these materials are readily available at low cost, leaf mold is not extensively used in container production.
Sawdust – the species of tree from which sawdust is derived largely determines its quality and value for use in a growing media. Several sawdusts, such as walnut and non-composted redwood, are known to have direct phytotoxic effects. However, the C:N of sawdust is such that it is not readily decomposed. The high cellulose and lignin content along with insufficient N supplies creates depletion problems which can severely restrict plant growth. However supplemental appli-cations of nitrogen can reduce this problem.

Barks – are primarily a bi-product of the pulp, paper and plywood industries. Suitable particle size is obtained by hammer milling and screening. This produces a material which is suitable for use in container media. Physical properties obtained from tree barks are similar to those of Sphagnum moss.

Bagasse

Bagasse is a waste bi-product of the sugar industry. It may be shredded and/or composted to produce a material which can increase the aeration and drainage properties of container media. Because of its high sugar content, rapid microbial activity results after the incorporation of bagasse into a media. This decreases the durability and longevity of bagasse and influences N levels. Although bagasse is readily available at low cost, (usually transportation), its use is limited.

Rice Hulls

Rice hulls are a biproduct of the rice milling industry. Although they are extremely light in weight, rice hulls are very effective at improving drainage. The particle size and resistance to decomposition of rice hulls and sawdust are very similar. However N depletion is not as serious of a problem in media amended with rice hulls.

Several other organic materials are suitable for use with container media. Included are: manures; corn cobs; straw; peanut and pecan shells. However these do not constitute major commercial sources of organic amendments.

Sand

Sand, a basic component of soil, ranges in particle size from 0.05mm to 2.0mm in diameter. Fine sands (0.05mm – 0.25mm) do little to improve the physical properties of a growing media and may result in reduced drainage and aeration. Medium and coarse sand particles are those which provide optimum adjustments in media texture. Although sand is generally the least expensive of all inorganic amendments it is also the heaviest. This may result in prohibitive transportation costs. Sand is a valuable amendment for both potting and propagation media.

Perlite

Perlite is a silicous mineral of volcanic origin. The grades used in container media are first crushed and then heated until the vaporization of combined water expands it to a light powdery substance. Lightness and uniformity make perlite very useful for increasing aeration and drainage.

Perlite is very dusty when dry and has a tendency to float to the top of a container during irrigation. It has also been shown that perlite contains potentially toxic levels of fluorine. Although costs are moderate, perlite is an effective amendment for growing media.

Vermiculite
Vermiculite is a micacious mineral produced by heating to approximately 745°C. The expanded, plate-like particles which are formed have a very high water holding capacity and aid in aeration and drainage. Vermiculite has excellent ex-change and buffering capacities as well as the ability to supply potassium and magnesium. Although vermiculite is less du-rable than sand and perlite, its chemical and physical properties are very desirable for container media.

Calcined Clays
Calcined clays are formed by heating montmorrillonitic clay minerals to aproximately 690°C. The pottery-like particles formed are six times as heavy as perlite. Calcined clays have a relatively high cation exchange as well as water holding capacity. This material is a very durable and useful amendment.

These inorganic soil amendments are generally utilized to increase the number of large pores, decrease water holding capacity and improve drainage and aeration. Other materials such as: pumice; cinders; and pea-gravel are also suitable for this use.

Several synthetic soil amendments are bi-products of various plastic manufacturing companies. Others are designed specifically for use in container media. These materials are frequently used in place of sand and perlite and have much the same influence on media properties.

Expanded Polystyrene

Polystyrene flakes, a bi-product of polystyrene processing, are highly resistant to decomposition, increase aeration and drainage,and decrease bulk density. Polystyrene may be broken down by high temperatures and by certain chemical disinfecting agents.

Urea Formaldehydes

This material is prepared by mixing air with a liquid resin and allowing to cool. Urea formaldehyde foams have a greater water holding capacity than polystyrene but are similar in their influence on aeration and drainage. Raw materials are easily transported and are very effective amendments.

Preparing Soilless Growing Media

Although amendment combinations may vary, basic objectives in the preparation of a growing media are alike. An effective program should produce a growing media that is:

1. porus and well drained, yet retentive of sufficient moisture to meet the water requirements of plants between irrigations;
2. relatively low in soluble salts, but with an adequate exchange capacity to retain and supply the elements necessary for plant growth;
3. standardized and uniform with each batch to permit the use of standardized fertilization and irrigation programs for each successive crop;
4. free from harmful soil pests; pathogenic organisms, soil insects, nematodes and weed seeds
5. biologically and chemically stable following pasteurization; primarily free from organic matter that releases ammonia when it is subjected to heat or chemical treatments.

Since innumerable amendment combinations can produce a growing medium with these characteristics, it is important to consider both the economic as well as cultural optimums. Factors that determine the cost of a growing medium include: transportation, labor, equipment, materials and handling. In many cases the cost of mixing a "custom" growing medium exceeds that of the commercially prepared materials. These factors should be studied carefully before making a decision.

Recommended Growing Media

The composition of a growing medium should be largely determined by the crop being produced. However there are some media formulations which may be used as a base. The following is a list of several of the most commonly used soilless mixtures:

Table 1. Commonly used soilless mixtures for greenhouse crops.

Volume/Volume Ratio	**Components**
2:1	Peat, Perlite[1]
2:1:1	Peat, Perlite, Vermiculite
2:1	Peat, Sand 3:1 Peat, Sand
3:1:1	Peat, Perlite, Vermiculite
2:1:1	Peat, Bark, Sand
2:1:1	Peat, Bark, Perlite
3:1:1	Peat, Bark, Sand

1. Foam beads may be used in place of perlite.

SELF CHECK 2.1-4

Prepare a Growing Media

1. ________________ What is a basic component of soil, ranges in particle size from 0.05mm to 2.0mm in diameter?

2. _______________ What is a siliceous mineral of volcanic?

3. ________________ What is a micaceous mineral produced by heating to approximately 745°C?

4. ________________What is a waste bi-product of the sugar industry. It may be shredded and/or composted to produce a material which can increase the aeration and drainage properties of container media?

5. ________________What is the dehydrated remains of acid-bog plants from the genus Sphagnum (i.e. Spapillosum)?

ANSWER KEY 2.1-4

Prepare a Growing Media

1. Sand
2. Perlite
3. Vermiculite
4. Bagasse
5. Sphagnum moss

JOB SHEET 2.1-4

Prepare a Growing Media

Title: Preparing the growing media
Performance Objective: At the end of this module you should be able to prepare the different kinds of growing media.
Supplies/Materials: PPE, farm, stocks, different growing media (carbonized rice hull, animal manure, compost)
Steps/Procedures: Wear the PPE. Gather the tools and materials Visit a farm Gather the available growing media Mixing all the growing media. Do housekeeping.
Assessment method: **Demonstration:**

PERFORMANCE CHECKLIST 2.1-4

Prepare a Growing Media

Performance Criteria	YES	NO
Did the trainee ..		
Wear the PPE?		
Gather the tools and materials?		
Visit a farm?		
Gather all the available growing media?		
Mixing all the growing media?		
Do housekeeping?		

Learning Outcome #2 PLANT SEEDLINGS

Assessment Criteria:

Land preparation is carried out in accordance with enterprise practice

Beneficial micro-organisms are introduced prior to planting in accordance with enterprise procedure

Seedlings are transplanted/planted based on VPM recommendations

Seedlings are watered based on VPM recommendations

Contents:
Land Preparation
Beneficial microorganisms
Planting/transplanting seedlings
Water seedlings

Tools, Materials and Equipment and Facilities

Farm/filed
Seedlings
Trowel
Sprinkler
water
PPE

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LEARNING EXPERIENCES/ACTIVITIES

LO # 2 PLANT SEEDLINGS

Learning Activities	Special Instructions
Read the attached Information Sheet 2.2-1 on Land Preparation	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 2.2-1 on Land Preparation	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 2.2-1 on Land Preparation	
Perform Job 2.2-1 on Land Preparation	Observe the demonstration of your trainer and check the procedures while performing the operation.
Evaluate performance using performance checklist on 2.2-1 on Land Preparation	
Read the attached Information Sheet 2.2-2 Beneficial microorganisms	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 2.2-2 Beneficial microorganisms	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 2.2-2 Beneficial microorganisms	
Read the attached Information Sheet 2.2-3 on Planting/transplanting seedlings	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 2.2-3 on Planting/transplanting seedlings	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 2.2-3 on Planting/transplanting seedlings	
Perform Job Sheet 2.2-3 on Planting/transplanting seedlings	Observe the demonstration of your trainer and check the procedures while performing the operation.

Evaluate performance using performance 2.2-3 on Planting/transplanting seedlings	
Read the attached Information Sheet 2.2-4 on Water seedlings	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 2.2-4 on Water seedlings	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 2.2-4 on Water seedlings	
Perform Job Sheet 2.2-4 on Water seedlings	Observe the demonstration of your trainer and check the procedures while performing the operation.
Evaluate performance using performance 2.2-4 on Water seedlings	

After doing this activity, you are now ready to proceed to the next module on Perform Plant Care and Management.

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INFORMATION SHEET 2.2-1

Land Preparation

Learning Objective: After reading this information sheet, you should be able to perform land preparation.

Introduction:

Land Preparation or Tillage Practice is a very important practice to enhance good yield from crops grown. It is one of the measures used to control crop diseases and pest invasion.

The purpose of land preparation is to provide the best soil conditions which will enhance the successful establishment of the tissue culture plants. It is one of the measures used to control crop disease and pest invasion. Land preparation is also called as tillage practice.
Tillage practice – is the mechanical pulverization or manipulation of the soil to take about favorable conditions for the growth of crops. Tillage practices include all operations used for the function of modifying the soil characteristics. It costs about 30 % of the total cost of cultivation.
The objective of land preparation is to develop potential tree growth, survival, and uniformity of a crop about to be established (planted). Through proper land preparation, factors that limit tree growth are reduced. These factors will be the included: You should be aware of land preparation types for proper implementation.

Tillage for Soil Conversation:

Tillage is an important and the main tool for conservation of the land. As per definition, its primary function is to provide a favorable soil environment for the plant growth which is indirectly related to soil conservation. The effect of tillage on soil erosion is the purpose of its several effects on soil such as aggregation surface sealing infiltration and resistant to erosion, destruction of soil structure either by excessive tillage or tillage operation at improper soil moisture condition tends to raise the soil erosion, causing significant soil loss. To achieve a better result for soil conservation the following points must be considered for tillage operations.

Tillage Depth

A plowing depth in the 15-20 cm range is generally adequate , and there is seldom any advantage in going deeper. In fact, shallower plowing is often suggested for low rainfall areas like the Sahel to conserve moisture.

In som areas, tractor-drawn sub-soilers (long, narrow shanks that penetrate down to 60 cm)are used in an effort to break up deep hardpans (compacted layers.)Results are fair to poor, depending on the kind of hardpan, those consisting of a dense clay coat oftten re-cement themselves within a short time.

Zero Tillage Practice

Zero Tillage Practice otherwise called **no tillage** is a simplified form of minimum tillage. It involves only opening a narrow strip about 2 to 3 cm wide or hole in the ground for seed or seedling placement. Zero tillage is no pre- planting seedbed preparation. Weeds are taken care of with the use of herbicides and cutlass without disturbing the land. The crop is then planted directly without tilling or ploughing the soil, tis process is highly effective under lands where soil and water erosion are heavy.

Conventional Tillage

It is the sequence of operations traditionally or most generally used given geographic area to produce a given crop. The operations used vary considerably for different crops and in different regions.

In the past, conventional tillage included moldboard plowing, usually in the fall. Spring action included one or more passes with a disk harrow or field cultivator before planting. More recently, conventional tillage has changed to contain the use of a chisel plow instead of a moldboard plow, and newer combination tools are replacing chisel plows. These implements leave additional residue than moldboard plows, but often not enough to qualify as conservation tillage.

The soil surface following conventional tillage as practiced in the past was effectively free of plant residue. This was helpful with older planting equipment that had limited capability to plant into the residue. It also buried weed seed and disease- bearing crop and weed residue, thereby helping to reduce problems with weeds and plant disease before the advent of modern chemeicla control.

The name **clean tillage** is used for any system that leaves the soil surface more or less free of residue. A soil surface essentially free of rsidues can be achieved with other implements, especially following a crop such as a soybean that produces fragile, easy-to-cover residue.Removing all residues from the soil surfaces and disturbing the soil surface greatly increase the potential for soil erosion. The potential for water erosion is fewer in flat fields, but the potential fow wind erosion is high.Improved planters, seed quality, and herbicides have largely eliminated the want to practice clean tillage.

Minimum Tillage

It is the small manipulation of the soil. It is otherwise referred to as traditional tillage process. It is not as sophisticated and technical tillage. It involves the use of the cutlass to slash weeds and vegetation regrowth on the farm, the less manipulation is done with the how and rake.

Farmers with access to tractor or animal drawn tiilage equipment may overdo tillage, particularly through repeated harrowing to control sprouting weeds or break up clods.Killing one crop of weeds by stirring the soil stimulates another by moving other weed seeds closer to the soil surface. Excessive tillage stimulates the microbial breakdown of humus and may further destroy good soil physical condition by over- pulverizing the soil, the machinery , animal, and foot traffic compact the soil, impairing root growth and drainage. Tillage is seldom excessive when hand tools are used to arrange ground for the reference crops, because of the amount of labor it involve. Slash and burn and slash and mulch methods fall under zero tillage, as do method using particularly adapted mechanical planters to sow seed into the unplowed ground. The plow or plant system described above or plowing and planting in one tractor pass are examples of minimum tillage. The savings on equipment wear and fuel are benefits where tractors are used.

SELF CHECK 2.2-1

Land Preparation

1. _________What is otherwise referred to as traditional tillage process?
2. _________What is the sequence of operations traditionally or most generally used given geographic area to produce a given crop?
3. ________It is otherwise called no tillage?
4. __________ It is used for any system that leaves the soil surface more or less free of residue.
5. ___________ What is an important and the main tool for conservation of the land. As per definition, its primary function is to provide a favorable soil environment for the plant growth which is indirectly related to soil conservation?

ANSWER KEY 2.2-1

Land Preparation

1. Minimum Tillage
2. Conventional Tillage
3. Zero tillage Practice
4. Clean Tillage
5. Tillage

JOB SHEET 2.2-1

Land Preparation

Title: Performing Land Preparation
Performance Objective: At the end of this module the student should be able to perform land preparation
Supplies/Materials: PPE, trowel, shovel, plow, rake, farm
Steps/Procedures: Wear PPE Gather the tools and Materials Visit a Farm Do clearing the area Perform Ploughing Do housekeeping
Assessment Method: Demonstration

PERFORMANCE CHECKLIST 2.2-1

Land Preparation

Performance Criteria	YES	NO
Did the trainee ..		
1. Wear PPE?		
2.Gather tools and materials?		
3. Visit a Farm?		
4. Do clearing the area?		
5 Perform ploughing?		
6 Do housekeeping?		

INFORMATION SHEET 2.2-2

Beneficial Microorganisms

Learning Objective: After reading this information sheet, you should be able to identify the different types beneficial microorganism.

Introduction:

Soil Microbes And What They Do For Plants

Did you know that only 40 to 60% of the fertilizer we apply actually goes to the plant, the remaining is lost to run off into our waterways, volatilization to the air or is tied up in the soils. This is why soil health is such an imperative piece of plant health. Functional soil is a soil embedded with organic matter and soil microbes that work together to hold onto nutrients in the soil and convert nutrients locked in the soil.

Beneficial soil microbes form symbiotic relationships with the plant. In fact, the plant will exert as much as 30% of its energy to the root zone to make food for microbes. In return those microbes not only protect the plant from stress, but also feed the plant by converting and holding nutrients in the soil.

What are the Different types of Soil Microbes?
There are five different types of soil microbes: bacteria, actinomycetes, fungi, protozoa and nematodes. Each of these microbe types has a different job to boost soil and plant health.

Bacteria
Bacteria is the crucial workforce of soils. They are the final stage of breaking down nutrients and releasing them to the root zone for the plant. In fact, the Food and Agriculture Organization once said “Bacteria may well be the most valuable of life forms in the soil.”

Actinomycetes

Actinomycetes were once classified as fungi, and act similarly in the soil. However, some actinomycetes are predators and will harm the plant while others living in the soil can act as antibiotics for the plant.

Fungi

Like bacteria, fungi also lives in the rootzone and helps make nutrients available to plants. For example, *Mycorrhizae* is a fungi that facilitate water and nutrient uptake by the roots and plants to provide sugars, amino acids and other nutrients.

Protozoa

Protozoa are larger microbes that love to consume and be surrounded by bacteria. In fact, nutrients that are eaten by bacteria are released when protozoa in turn eat the bacteria.

Nematodes

Nematodes are microscopic worms that live around or inside the plant. Some nematodes are predators while others are beneficial, eating pathogenic nematodes and secreting nutrients to the plant.

SELF CHECK 2.2-2
Beneficial microorganisms

1. _______________ What are the microscopic worms that live around or inside the plant?
2. _______________ What are the larger microbes that love to consume and be surrounded by bacteria?

3. _______________What is a soil embedded with organic matter and soil microbes that work together to hold onto nutrients in the soil and convert nutrients locked in the soil?

4. ______________ What fungus that facilitate water and nutrient uptake by the roots and plants to provide sugars, amino acids and other nutrients?

5. _______________How many percent of fertilizer we apply actually goes to the plant?

ANSWER KEY 2.2-2
Beneficial microorganisms

1. Nematodes
2. Protozoa
3. Beneficial Soil
4. Mycorrhizae
5. 40 to 60%

INFORMATION SHEET 2.2-3

Planting/transplanting seedlings

Learning Objective: After reading this information sheet, you should be able to plant/transplant vegetable seedlings.

Introduction:

Whether you started seeds yourself or purchased vegetable seedlings at the store, now is the time to transplant seedlings in your garden.

10 simple steps to transplant

1. Seedlings should be hardened-off, well-fed and watered before transplanting.
2. Prepare a weed-free surface. Loosen and aerate garden soil by tilling or hoeing.
3. Dig a hole large enough for seedling.
4. Carefully remove seedling from its container. Try not to disturb the roots.
5. Set seedling in hole level with soil surface. The exception is tomato seedlings, which can be transplanted a bit deeper.
6. Feed seedling to kick start growth. I transplant each seedling with a hefty handful of compost. If you don't make compost, purchase specially formulated fertilizer for transplanting.
7. Surround seedling with displaced soil.
8. Water seedling thoroughly.
9. Mulch seedling to maintain soil moisture and regulate temperature.
10. Keep area weed-free.

Top 3 tips for successful transplanting

Start with strong seedlings. If you start your own seeds, make sure your seedlings have plenty of light, adequate water and drainage. A good growing medium is critical for root establishment.

If you buy seedlings remember biggest doesn't always mean best. Look for healthy and consistent leaf color. The roots should be deep, long, white and fibrous. The stem should be thick and strong. The growing medium should be held together by tight tangled roots. Do not choose leggy seedlings; too long of stem means the seedling was starved for light.

Hardening off is important. Don't skip this step. Store bought seedlings are typically hardened off in the garden center, but seedlings you start need to be acclimated to the natural environment before transplanting.

Hardening takes 1-2 weeks. You can move seedlings to a cold frame for a week and then set in the garden an additional week. Or set seedlings out during warm daytime temps and bring them in at night.

Timing is everything. Frost-free dates and warm and cool weather crop temperature requirements for your growing zone are good guidelines. I also consider last season's planting dates and the weather forecast.

Overcast days are great days to plant because cloud coverage reduces the probability of sun-scorching tender plants.

SELF CHECK 2.2-3
Planting/transplanting seedlings

TRUE OR FALSE

1. ____________________ Seedlings should be hardened-off, well-fed and watered before transplanting.

2.____________________ Prepare a weed-free surface. Loosen and aerate garden soil by tilling or hoeing.

3.____________________ Dig a hole large enough for seedling.

4.____________________ Carefully remove seedling from its container. Try not to disturb the roots.

5.____________________ Water seedling thoroughly.

ANSWER KEY 2.2-3

Planting/transplanting seedlings

1 True

2. True
3. True
4. True
5. True

JOB SHEET 2.2-3

Planting/transplanting seedlings

Title: Planting/Transplanting Vegetable Seedlings
Performance Objective: At the end of this module the student should be able to transplant vegetable seedlings.
Supplies/Materials: PPE, Trowel, Seedlings, field
Steps/Procedures: Wear the PPE. Prepare the tools and materials Go to the Field. Measure the proper planting distance of vegetables Make hole for seedlings. Do transplanting of seedlings
Assessment Method: Demonstration

PERFORMANCE CHECKLIST 2.2-3

Planting/transplanting seedlings

Performance Criteria	YES	NO
Did the trainee ..		
1. Wear the PPE?		
2. Prepare the tools and materials?		
3. Go to the Field?		
4. Measure the proper planting distance of vegetables?		
5. Make hole for seedling?		
6. Do transplanting of seedlings?		

INFORMATION SHEET 2.2-4

Water Seedlings

Learning Objective: After reading this information sheet, you should be able to water seedlings.

Introduction:

Water is one of the vital elements when starting plants from seed. Too much water and your seeds will drown or rot. Too little and they will either fail to germinate or die once they do.

If you are starting your vegetable garden from seed, you have two choices. One, you can start your seeds indoors and then plant them outside as seedlings several weeks later, or you can direct seed into your garden.

There are a number of good reasons to start seeds early indoors. Most importantly, you get ahead of the growing season. This is especially important if you live in a place with a short growing season. Another advantage is that you can tightly control the ideal growing conditions: temperature, moisture, sunlight, etc. A third advantage is cooling that early spring itch to get outside and get something in the ground!

The best candidates for early starts are things like broccoli, cabbage, cauliflower, celery, eggplant, leeks, onions, parsley, peppers, and tomatoes. Root crops, like beets and carrots do not like to be transplanted and are best sown directly into the garden. Corn and peas are other things that do not take well to a transplanting.

To germinate seeds you can do one of two things:

1. You can moisten a paper towel, place the seeds in the middle of the paper towel and place it on the window sill in the sunlight.

2. Alternately, you can fill small sections of a seed starting tray with a soil mixture and plant the seeds into the mixture about 1 inch deep. Then water lightly.

Either way, you don't want the seeds sitting in water. You want to have the soil or paper towel moist but not soaked. Let the soil mix dry out just a bit, but not completely, before wetting again. I use a spray bottle to keep my starting mixture moist. The key is good drainage. Make sure that excess water has a way to drain away from the seeds. There are seed starting systems available that work via a capillary system. This keeps your soil at the right moisture level with no work from you but I find the spray bottle method pretty simple.

If you cover your seeds with some loose plastic, you will create a mini-greenhouse environment that will hold in both heat and moisture. You will need to get air to the seeds so remove the plastic every once in a while so that you don't get mold formation that can ruin your seeds.

After a few days, two small leaves will appear once your seedlings begin poking through the soil and unfurling. Again, keep the soil moist as the seedling begins to take off

If you cover your seeds with some loose plastic, you will create a mini-greenhouse environment that will hold in both heat and moisture. You will need to get air to the seeds so remove the plastic every once in a while so that you don't get mold formation that can ruin your seeds.

After a few days, two small leaves will appear once your seedlings begin poking through the soil and unfurling. Again, keep the soil moist as the seedling begins to take off.

As the days warm and lengthen you will begin taking your seedlings outdoors to "harden off". That is, to get them used to being outside by putting them out for portions of the day (you will still be bringing them inside at night). Be careful here. The sun and spring winds can dry out that delicate soil in a heartbeat.

When the day comes that you are ready to plant your seedlings into the garden, water them well before the transplant. Once they are in the garden, water them again very well. Finally, to avoid drying out your seedlings try not to transplant during the hottest, sunniest part of the day.

SELF CHECK 2.2-4

Water Seedlings

TRUE OR FALSE

1. ____________________ Water is one of the vital elements when starting plants from seed.
2. ____________________ Too much water and your seeds will drown or rot.

3. ____________________ Fill small sections of a seed starting tray with a soil mixture and plant the seeds into the mixture about 1 inch deep.

4. _____________________ There are a number of good reasons to start seeds early indoors.

5. ____________________ Make sure that excess water has a way to drain away from the seeds.

ANSWER KEY 2.2-4

Water Seedlings

1. TRUE
2. TRUE
3. TRUE
4. TRUE
5. TRUE

JOB SHEET 2.2-4

Water Seedlings

Title: Watering Seedlings
Performance Objective: At the end of this module the student should be able to water seedlings.
Supplies/Materials: PPE, container, sprinkler, field, newly planted seedlings
Steps/Procedures: Wear the PPE. Prepare the tools and materials Go to the Field. Pitch water from the water supply Do watering on seedlings. Do house keeping
Assessment Method: Demonstration

PERFORMANCE CHECKLIST 2.2-4

Water Seed and Seedlings

Performance Criteria	YES	NO
Did the trainee ..		
1. Wear the PPE?		
2. Prepare the tools and materials?		
3. Go to the Field?		
4. Pitch water from water supply?		
5. Do watering on seedlings?		
6. Do housekeeping?		

Learning Outcome #3 Perform Plant care and Management

Assessment Criteria:

Water management is implemented according to plan.

Effective ***control measures*** are determined on specific pest and diseases as described under the "pest, disease and weed management" of the PNS.

All missing hills are replanted to maintain the desired plant population of the area.

Plant rejuvenation/rationing are maintained according to PNS.

Organic fertilizers are applied in accordance with fertilization policy of the PNS

Contents:

Water management implementation
Pest and diseases control measures
Replanting missing hills
Plant rationing
Organic fertilizer application

Tools, Materials and Equipment and Facilities

Seedlings

Organic fertilizer (compost, animal manure)

botanical repellants

hose

water source

References:

https://www.un.org/waterforlifedecade/iwrm.shtml

https://www.growpittsburgh.org/garden-and-farm-resources/info-hub/pest-disease-management/

https://www.hortweek.com/plant-rationing-comes-sales-fly/ornamentals/article/1710204

https://ph.search.yahoo.com/yhs/search?hspart=trp&hsimp=yhs-001&type=64891_070717&p=organic+fertilizer+application

LEARNING EXPERIENCES/ACTIVITIES

LO # 3 Perform Plant care and Management

Learning Activities	Special Instructions
Read the attached Information Sheet 2.3-1 on Water management implementation	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 2.3-1 on Water management implementation	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 2.3-1 on Water management implementation	
Perform Job 2.3-1 on Water management implementation	Observe the demonstration of your trainer and check the procedures while performing the operation.
Evaluate performance using performance checklist 2.3-1 on Water management implementation	
Read the attached Information Sheet 2.3-2 on Pest and diseases control measures	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 2.3-2 on Pest and diseases control measures	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 2.3-2 on Pest and diseases control measures	
Perform Job Sheet 2.3-2 on Pest and diseases control measures	Observe the demonstration of your trainer and check the procedures while performing the operation.
Evaluate performance using performance checklist 2.3-2 on Pest and diseases control measures	

Read the attached Information Sheet 2.3-3 on Replanting missing hills	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 2.3-3 on Replanting missing hills	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 2.3-3 on Replanting missing hills	
Perform Job Sheet 2.3-3 on Replanting missing hills	Observe the demonstration of your trainer and check the procedures while performing the operation.
Evaluate performance using performance checklist 2.3-3 on Replanting missing hills	
Read the attached Information Sheet 2.3-4 on Plant rationing	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 2.3-4 on Plant rationing	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 2.3-4 on Plant rationing	
Perform Job Sheet 2.3-4 on Replanting missing hills	Observe the demonstration of your trainer and check the procedures while performing the operation.
Evaluate performance using performance checklist 2.3-4 on Replanting missing hills	
Read the attached Information Sheet 2.3-5 on Organic fertilizer application	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 2.3-5 on Organic fertilizer application	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 2.3-5 on Organic fertilizer application	

After doing this activity, you are now ready to proceed to the next module on Perform harvest and post-harvest activities.

INFORMATION SHEET 2.3-1

Water management implementation

Learning Objective: After reading this information sheet, you should be able to appreciate the importance of proper water management.

Introduction

Why Is Irrigation Water Management Important?

Irrigation water management is the act of timing and regulating irrigation water application in a way that will satisfy the water requirement of the crop without wasting water, energy, and plant nutrients or degrading the soil resource. This involves applying water according to crop needs in amounts that can be held in the soil and at rates consistent with the intake characteristics of the soil.

A primary objective in the field of irrigation water management is to give irrigators an understanding of conservation irrigation principles. This is done by showing them how they can judge the effectiveness of their own irrigation practices, make good water management decisions, or recognize the need to make adjustments in existing systems or to install new systems. The net result of proper irrigation water management typically:

- Prevents excessive use of water
- Minimizes pumping costs
- Prevents excessive soil erosion
- Reduces labour
- Maintains or improves quality of groundwater and downstream surface water
- **Increases crop biomass yield and product quality**

Irrigation scheduling is the part of proper irrigation water management that involves the decision of when to irrigate and how much water to apply. Scheduling tools provide information that irrigation decision makers can use to develop irrigation strategies for each field on the farm. Such strategies may be based on long-term data that represents average conditions or may be developed as the season progresses, using real-time information and short-time predictions. In both cases, information about the crop, soil, climate, irrigation system, water deliveries, and management objectives must be considered to tailor irrigation scheduling procedures to a specific irrigation decision maker and field condition. An irrigation scheduling tool needs only be accurate enough to make the decision when and how much to irrigate.

Modern scheduling is based on soil-water balance or crop-water balance for one or more points in the field. By measuring existing and estimating future soil-water content or by monitoring crop-water stress level, irrigation water can be applied before damaging crop stress occurs. Scheduling irrigation involves forecasting of crop water use rates to anticipate future water needs.

Computerized irrigation scheduling allows the storage and transfer of data, easy access to data, and calculations using the most advanced and complex methods for predicting crop evapotranspiration. Many computer software programs are available to assist in scheduling irrigations. In Kansas, software available from the Kansas State Research and Extension is typically used. The software (called KanSched) is used by the Natural Resources Conservation Service (NRCS) to document irrigation water management under the Environmental Quality Incentives Program.

When you are a farmer, gardener or work in an industry that requires landscape maintenance it's important to implement a system of irrigation management. In order to produce proper crops or fields of flowers, healthy soil and plants must be watered routinely at the appropriate times and rates. What's more, proper rates of water and timing are essential for properly yielded results. Oftentimes systems are too extensive or the size of properties to vast to water areas manually which is why automatic systems are used more readily for convenience and ease of use. Be sure to get in touch with professional irrigation management services whenever you are curious as to how to go about securing a system. If you're in need of landscaping services or whole house water treatment be sure to read the following before you get in touch with a professional.

Why do we use irrigation?

Irrigation management is essential for gardeners or farmers in order to promote plant growth. In turn, stock farmers can use them in order to make sure their animals have available food sources for healthier systems. Since crops and plants need routine moisture the use of irrigation systems has risen in their essential need.

What is irrigation water management?

Water irrigation management involves the monitoring of water application for crops or yard. It usually will be used for more extensive properties that need a system to help manage the volume, rate, and timing of water application in order to match with water holding capacities and soil intake. In order to promote optimum crop yields, it's especially important to monitor soil moisture without runoff or deep percolation losses. With your irrigation management, you'll be able to properly adjust your water with tools that can, later on, be adjusted to ensure properly yielded results. Flow meters are great at recording instantaneous flow rates as well as the total volume of water used. With soil moisture sensors and meters soil water deficit may be monitored. A checkbook method, in turn, may be able to balance soil moisture through its monitoring of leveling an irrigated cropland. Finally, with data loggers, you can record soil moisture history through the growing season of a field or area.

What are the objectives of irrigation?

The main objectives for irrigation management or irrigation, in general, is to promote the proper growth of plants and maintaining the right levels of moisture for the soil. Another objective can be seen as making sure there is backup insurance when there is a short duration of drought as this can sustain the field enough when water levels are low. Another reason is to cool the atmosphere and soil which is an ideal environment for plants. What's more, with regular irrigation you'll be able to dilute any harsh chemicals in soil or even washout harmful salts. Finally, with irrigation you can lower the dangers of soil piping which can increase subsurface erosion from the unnatural underground water flow. Soil piping is an alternate method from irrigation yet it can threaten farming results as well as threaten the stability of any surrounding buildings with movement when soil moisture levels are too high and cause the building to shift.

What are the methods of irrigation?

There are various methods that can be applied to your irrigation management system. Irrigation management methods will be dependent on what your irrigation is being used for or its goals.

The Different Irrigation Methods Used

- Center Pivot Irrigation
- Drip Irrigation
- Lateral Move Irrigation
- Localized Irrigation
- Manual Irrigation
- Sprinkler Irrigation
- Sub-Irrigation
- Surface Irrigation

What is the best method of irrigation?

The best typed of irrigation management for you may depend on the specifics of what is needed as well as the characteristics of your property. In general, however, it seems that drip irrigation is one of the most efficient types of irrigation systems due to their percentages of applied and lost water ratings in conjunction with meeting crop water need falling between 80-90%. It’s important to take into consideration irrigation management and problems that may befall your particular property with a professional in order to ensure the right method is applied for you.

SELF CHECK 2.3-1

Water management implementation

1. ________________ The main objectives for irrigation management or irrigation, in general, is to promote the proper growth of plants and maintaining the right levels of moisture for the soil.

2. ________________ Irrigation water management is the act of timing and regulating irrigation water application in a way that will satisfy the water requirement of the crop without wasting water, energy, and plant nutrients or degrading the soil resource.

3. ________________Water irrigation management involves the monitoring of water application for crops or yard.

4. ________________ drip irrigation is one of the most efficient types of irrigation systems due to their percentages of applied and lost water ratings in conjunction with meeting crop water need falling between 80-90%.

5. ________________ Irrigation management is essential for gardeners or farmers in order to promote plant growth.

ANSWER KEY 2.3-1

Water management implementation

1. True
2. True
3. True
4. True
5. True

JOB SHEET 2.3-1

Water management implementation

Title: Implementing water management
Performance Objective: At the end of this module you should be able to implement water management.
Supplies/Materials: Sufficient Water source , hose, shovel
Steps/Procedures: Wear the PPE. Gather the tools and materials. Go to the Field. Make a drainage canal along the plot. Install the hose. Do housekeeping.
Assessment method: **Demonstration:**

PERFORMANCE CHECKLIST 2.3-1

Water management implementation

Performance Criteria	YES	NO
Did the trainee ..		
Wear the PPE?		
Gather the tools and materials?		
Go to the field?		
Make canal along the fields?		
Install the hose?		
Do housekeeping?		

INFORMATION SHEET 2.3-2

Pest and diseases control measures

Learning Objective: After reading this information sheet, you should be able to apply control measures on pest and diseases.

Introduction:

Once a pest has reached either an economic threshold or an intolerable level, action should be taken. Pesticides are used as a control measure when other strategies will not bring the pest population under the threshold, when other strategies are too expensive or time-consuming, or when the quality or yield effects are unacceptable to the grower. In fact, the success of waiting until a pest reaches the threshold usually hinges on the availability of a pesticide that will bring the pest populations down quickly.

Management tactics

Management tactics can be preventative, curative, or both and are sometimes combined to provide the best possible program. Preventative measures, taken before planting, or before the pest appears, can result in fewer rescue treatments. Each crop and situation will require management options tailored to that situation. A general list of actions is provided below.

Cultural Controls are those that disrupt the environment of the pest, and/or prevent its movement. Plowing, crop rotation, removal of infected plant material, cleaning of greenhouse and tillage equipment, and effective manure management are all cultural practices that are employed to deprive pests of a comfortable habitat or prevent their spread. The management of urban and industrial pests has improved with proper sanitation and elimination of pest harborages, more frequent garbage pickup, or installation of lights that do not attract insects.

- Rotate crops to reduce the buildup of weeds, disease, and insect pests. Crop rotation is useful for those pests that do not move far from their overwintering sites.
- Remove overwintering sites, such as cull piles, damaged, and volunteer plants, and alternate hosts, to minimize damage by insects and diseases.

- Use techniques that expose pests to natural enemies or environmental stress, or that make the crop less susceptible to insects or diseases.
- Adjust planting times to avoid periods of peak pest abundance.
- Plant disease-free seeds and transplants.
- Promote vigorous crop growth with proper nutrition and weed removal to avoid stress that may weaken crops and make them more susceptible to attack by insects, diseases, or physiological disorders.
- Manage irrigation schedules to avoid long periods of high relative humidity. Wet, highly humid conditions encourage disease pests to develop. When possible only irrigate the root system and not the foliage.
- Arrange fields for the best air drainage and circulation to promote low humidity.
- Where crops are planted in rows use cultivation, where practical, in combination with banding of herbicides over the row for weed control. This could reduce herbicide costs while achieving good weed control.
- Keep woody plants thinned to improve air circulation within the plant foliage.
- Use a no-till system to reduce weed seed germination.
- Plant cover crops to prevent weeds from germinating after harvest.

Physical Barriers such as netting over small fruits and screening in greenhouses can prevent insects that cause crop loss, and mulch can inhibit weed germination beneath desirable plants. Physical barriers are important in termite, house fly, and rodent control. Paint or seal porous wood.

Biological Controls — conserving or releasing natural enemies (biological control agents) can prevent the rise of certain pests. Examples of biological control agents are beneficial mites that feed on mite pests in orchards, the *Hb* nematodes that kill harmful soil grubs, and *Encarsia formosa*, a wasp that parasitizes the greenhouse whitefly. Many biological control agents are commercially available.

- Purchasing and releasing predators and parasites of pests, if available, can be effective in reducing pest populations especially in greenhouses or other enclosed structures.
- Develop refuges for natural enemies of the pest by establishing areas of flowering plants and shrubs to supply nectar, alternative hosts, and shelter.
- Choose and use pesticides wisely so you can conserve indigenous or released natural enemies of insect and mite pests.

Pest-resistant cultivars are less susceptible than other varieties to certain insects and diseases. Planting, disease-resistant crops is one of the simplest methods of reducing disease management actions during the growing season. The use of resistant varieties often means that growers need not apply as many pesticides as with susceptible varieties. Potato growers control the golden nematode by planting resistant cultivars. Apple growers can save up to eight fungicide applications a year by growing certain cultivars that resist diseases. Farmers growing alfalfa and wheat keep several pests at bay by planting resistant varieties. Many ornamental plant cultivars have been bred to resist diseases and insects. American elm and American chestnut may return to our forests in the future as the result of genetically modified cultivars.

Pesticides

- One of the major goals of IPM is to minimize reliance on pesticides.
- Use pesticides only when monitoring, economic thresholds, or disease forecasts indicate a need and with the appropriate timing, on target, and at the lowest effective rate.
- Select pesticides that are registered by both the state and EPA and labeled for use on the intended crop or site. Also select according to efficacy, previous use patterns, the potential for and incidence of resistance, and the possible impact on the environment and natural enemies.
- Be certain to achieve uniform coverage with your equipment, applying recommended application rates with accurately calibrated equipment that targets the pest, or crop surfaces to be protected.

Visit other sections of the PES site to learn about proper stewardship concerning pesticide use.

Structural Modifications, such as preventing support timbers from contacting soil, can help prevent damage from several different wood-destroying pests. Wood absorbs moisture and is more susceptible to attack by carpenter ants and termites when in direct contact with the soil. Paints, sealants, or other barrier applications can also prevent pest intrusion into structural materials.

Construction Site Sanitation, such as removing tree stumps and lumber scraps from construction sites, which are prime food sources for subterranean termites, can prevent problems in the future.

SELF CHECK 2.3-2

Pest and diseases control measures

TRUE OR FALSE

1. ________________ Rotate crops to reduce the buildup of weeds, disease, and insect pests. Crop rotation is useful for those pests that do not move far from their overwintering sites.

2. ________________ Remove overwintering sites, such as cull piles, damaged, and volunteer plants, and alternate hosts, to minimize damage by insects and diseases.
3. Use pesticides only when monitoring, economic thresholds, or disease forecasts indicate a need and with the appropriate timing, on target, and at the lowest effective rate.
4. ________________ Arrange fields for the best air drainage and circulation to promote low humidity.
5. ________________ One of the major goals of IPM is to minimize reliance on pesticides.

ANSWER KEY 2.3-2

Pest and diseases control measures

1. True
2. True
3. True
4. True
5. True

JOB SHEET 2.3-2

Pest and diseases control measures

Title: Controlling pest and diseases
Performance Objective: At the end of this module you should be able to control pest and diseases.
Supplies/Materials: PPE, botanical repellant, sprayer and water, field
Steps/Procedures: Wear the PPE. Prepare botanical repellant Put some botanical repellant into a sprayer of water. Apply it into the filed Do housekeeping
Assessment method: **Demonstration:**

PERFORMANCE CHECKLIST 2.3-2

Pest and diseases control measures

Performance Criteria	YES	NO
Did the trainee ..		
wear the PPE?		
Prepare botanical repellant		
Put some botanical repellant into a sprayer of water?		
Apply it into the filed?		
Do housekeeping?		

INFORMATION SHEET 2.3-3
Replanting missing hills

Learning Objective: After reading this information sheet, you should be able to do replanting.

Introduction:

Replanting when crop damage and stand reduction occurs early in the growing season can be an economically viable option. Replanting decisions, however, are complicated by not knowing what future seasonal growing conditions will occur. Decisions should be based on historic weather trends plus current environmental and economic conditions.

The decision to replant should be made only after evaluating the following questions: Will the economic returns exceed the cost of replanting? What is the most viable crop to plant? Furthermore, carefully consider how current soil moisture, previous herbicide use and the date of replanting might influence the crop or crops to be replanted.

The final decision on replanting should be based on sound agronomic and economic information. Injury to the original stand, as well as crop uniformity and overall plant health, must be determined accurately. The initial critical question is whether keeping the original stand or replanting will result in greater net income.

Late Planting

Excessive moisture, poorly drained soils and other factors frequently delay planting beyond the optimum period for yield in North Dakota. Since many of the factors that impact the decision on replanting also impact the decision on how to deal with late planting, this publication also provides relevant information on delayed crop planting

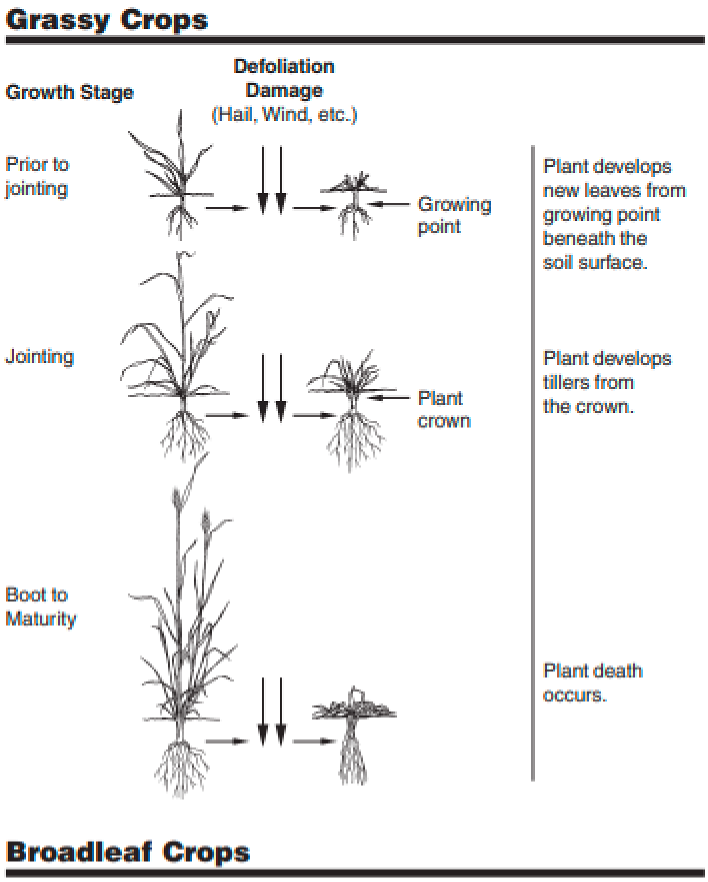
Appraising Crop Injury, Stand Reduction and Yield Potential

Evaluating crop injury and estimating potential crop yield is the first step in determining if a crop should be replanted. The best possible evaluation of the surviving stand is needed because the critical yield comparison ultimately will be between the suboptimal stand from the original planting date versus a full stand from a later than optimum planting date.

During the seedling stage, injury that results in stand reduction will cause the greatest yield reductions. Leaf loss or leaf burn during early stages has a minimal effect on yield; however, as the crop approaches reproductive developmental stages, leaf damage or loss is more detrimental to yield. An assessment of potential plant stand soon after crop injury occurs is critical to avoid delays if replanting is necessary.

An accurate determination of the existing stand must be made. Stand counts should be taken at random from several areas in damaged portions of the field. Determining stand level may be more complicated than simply counting plants. If seedling emergence is uneven, an evaluation of potential late-emerging plants should be made. If areas in a field are not damaged, they do not need to be considered for replanting. However damage in fields often is distributed randomly throughout a field and this complicates decision making.

In early plant development, a few key structures are indicators of potential plant survival. A healthy root system and seedling emergence tissue are needed during plant establishment ***(see Figure 1 for key structures)***. If these structures appear normal and the depth of the seed is not excessive, emergence likely will occur when soil moisture is adequate. Destruction of leaf area on young plants is seldom



as detrimental to subsequent plant growth and yield as the initial appearance may suggest. If the growing point of the small grain is not damaged and the stem is not broken below the cotyledonary node of broadleaf crops, the plants likely will recover.

During early growth stages, most North Dakota crops can sustain some stand loss without experiencing significant yield reduction. Crops compensate for stand reduction through tillering, secondary branching or increased head number, and increased ear, head or seed size.

Comparison of the estimated yield of the injured crop with expected yield of an alternative crop minus reseeding costs must be considered. Often this calculation will reveal that the present crop stand will be more viable economically than a later replanted crop.

Table 1 shows the population levels of several crops that should be considered minimum stands when deciding whether to replant, assuming the plant population is relatively uniform in distribution. As the season progresses, the yield potential of a newly planted crop is reduced **(Tables 2 and 3)**; therefore, the lower minimum stand values from **Table 1** should be utilized as the season progresses.

During early development, the growing point of cereal crops is below the soil surface, making them less susceptible to injury. With this protection, plants can suffer the loss of above-ground foliage without dying.

When hail, frost or similar types of injury cause severe foliar damage, waiting several days after the injury occurs to make an accurate determination of stand reduction is advisable. After this period, new growth on plants with uninjured growing points can be observed as in **Figure 1**. If no regrowth is observed, the stem of the plant may be cut in half to inspect the growing point.

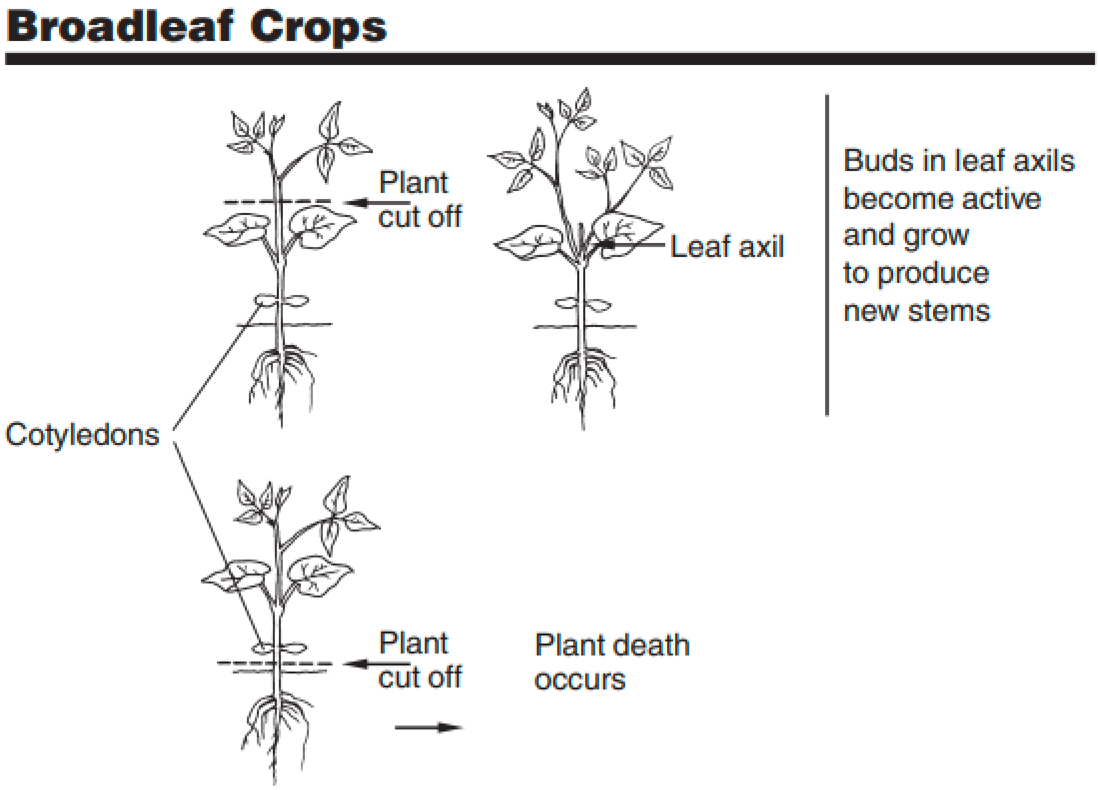


Figure 1. Effect of injury on grassy crops at different growth stages and on broadleaf crops.

Table 1. Minimum stand levels of several crops that should be considered acceptable before replanting is done.

Crop	% of Normal Stand	Minimum Stand	Units/Notes
Canola	25-50	4	plants/ft^2
Corn: Irrigated	50-60	17,000-20,000	plants/A
Corn: Dryland	40-50	13,000-16,000	plants/A
Dry Bean: Navy and Black	50-60	45,000-60,000	plants/A
Dry Bean: Pinto	40-50	28,000-35,000	plants/A
Field pea	40-70	3-5	plants/ft^2
Flax	20-40	20-35	plants/ft^2
Safflower	40-50	2-2.5	plants/ft^2
Small Grain	30-40	8-14	plants/ft^2
Soybean	50	75,000	plants/A
Sunflower	50-60	8,000-11,000	plants/A

Table 2. Expected yield reductions when planting cool-season crops late in North Dakota.

Crop	Yield loss/day
	(%)[a]
Barley	1.7
Wheat	1.5
Oat	1.2
Canola	1.9
Field Pea	2.1
Flax	2.3

[a] Percent of optimal planting date.

Table 3. Expected yield reductions when planting corn, soybean and sunflower late in North Dakota.

Crop	Yield loss/day
	(%)[a]
Corn[b]	2.0
Soybean[c]	0.6
Sunflower[c]	1.8

[a] Percent of optimal planting date.
[b] Planting after May 25.
[c] Planting after May 31.

The growing point should be white or cream-colored. Darkening or softening of the growing point usually precedes plant death. When the growing point moves above the soil surface at jointing in small grains and the sixth-leaf stage in corn, it becomes more vulnerable to physical damage.

Cereal Crops

When hail severely injures small-grain cereal crops after jointing, plants still have potential for recovery by initiating new tillers. Precipitation that usually accompanies hail storms will help stimulate tillering. Potentially, tillering can restore yield potential to acceptable levels. The number of head-bearing tillers is determined before heading of the main stem, so injury that occurs after heading of the main stem in small grains is the most damaging to yield

Broadleaf Crops

The initial growing point on most broadleaf crops is at the plant's tip, which increases susceptibility to injury early in the season. Stem breakage or damage below the cotyledons (Figure 1) will result in plant death. However, buds form in the leaf axils later in the development of broadleaf plants, unlike small grains. After the loss of the main stem from injury, these buds can begin growth with secondary branching replacing the loss of the main stem. When this occurs, yield reduction will result from leaf loss only, rather than a combination of stand and leaf loss. The cotyledons of field pea and lentil remain under ground and the secondary growing points near the cotyledons can be activated if damage occurs to the top of the plant.

In sunflower, secondary stems developed from buds in the leaf axils will not compensate for loss of the main stem. These secondary stems will not produce viable seed heads, so loss of the main stem growing point results in the loss of the plant's yield potential. In addition, injured plants with secondary stems compete with healthy plants, further reducing yield.

Indirect Effects of Injury

In addition to the direct effect of leaf loss or stand reduction, indirect effects of crop injury, such as increased weed competition and increased disease potential, should be considered. Damaged crops usually grow slowly until they have recovered, which provides the potential for greater weed competition. Loss of leaf canopy allows additional sunlight to reach previously shaded weeds and may result in additional weed flushes. Wounds from hail, insect or wind injury provide opportunities for pathogens to infect the plant. Resulting diseases may reduce yields or grain quality directly.

Previously applied herbicides may have little remaining activity to control new flushes of weeds. Use of postemergence herbicides (particularly those that normally stress the crop) to control new weed flushes should be delayed on visibly weakened crops until the crop has an opportunity to recover. Otherwise, herbicides that are relatively safe on the crop should be used to control new weed flushes when weeds are likely to become too large to control once the crop has fully recovered.

Areas with no vegetative cover are in danger of soil erosion and stay wet longer because no plant transpiration is taking place.

Replanting Decision

The decision to replant ultimately must be made by comparing the estimated yield of the injured crop with that of a replanted crop. This is quite subjective and each case must be considered individually in terms of time of year, alternate crop choices, previous herbicide use, crop economics and insurance, and other related factors.

Crops replanted late in the season almost always will yield less than those planted at an optimum time. **Table 2** shows approximate yield reductions that may be expected from late-planted cool-season crops. **Table 3** shows approximate yield reductions that may be expected from late-planted corn, soybean and sunflower. The remaining growing season may be too short for some crops. **Table 4** shows options for replanting and the dates by which most North Dakota crops can be planted safely and still produce a useful or marketable product. These dates will vary somewhat depending on the region of the state.

Residues from previously applied herbicides may restrict some crops from being used as a replant crop.

Refer to herbicide labels and the current "North Dakota Weed Control Guide" (publication W-253) for information that indicates crops that, under normal conditions, can be planted safely on soils previously treated with one or more herbicides.

The decision to replant is both an agronomic and economic one that requires careful assessment of crop injury, yield potential, alternate crop choices and cultural practices related to crop growth and development. Each case of injury must be considered thoroughly and individually.

Table 4. Field crop replanting suggestions for North Dakota when soil moisture is not limitin

Plant Date	Barley	Buck-wheat	Canola	Corn	Dry Bean	Flax	Millet	Mustard	Pea, Lentil	Saf-flower	S be
May 1-10	1	–	1	–	–	1	–	1	1	1	[illegible]
May 11-20	1	–	1	1	1	1	–	1	1	3	[illegible]
May 21-31	1,5	1	3	3	1	3	–	3	3	4	[illegible]
June 1-10	2,4	1	4	2,4	3	3	1	4	4	4	[illegible]
June 11-20	2,4	1	4	2,4	3	4	1	4	4	4	[illegible]
June 21-30	2,4	1	4	4	4	4	3,4	4	4	4	[illegible]
>July 1	2,6	6	6	6	6	6	6	6	6	6	[illegible]

1. Replant.
2. Replant for use as silage or grazing forage.
3. Replant with earlier-maturing varieties/hybrids.
4. Use alternative crop with shorter maturity. (Be aware of possible herbicide residue restrictions).
5. Replant using higher seeding rates (1.3 million to 1.6 million plants/acre for small grain).
6. Do not replant.

SELF CHECK 2.3-3

Replanting missing hills

TRUE OF FALSE

1. ________________Replanting when crop damage and stand reduction occurs early in the growing season can be an economically viable option. ________________ When hail severely injures small-grain cereal crops after jointing, plants still have potential for recovery by initiating new tillers.
2. ________________ Evaluating crop injury and estimating potential crop yield is the first step in determining if a crop should be replanted.
3. ________________ Excessive moisture, poorly drained soils and other factors frequently delay planting beyond the optimum period for yield.
4. ________________ The decision to replant ultimately must be made by comparing the estimated yield of the injured crop with that of a replanted crop.
5. ________________ The final decision on replanting should be based on sound agronomic and economic information.

ANSWER KEY 2.3-3
Replanting missing hills

1. True
2. True
3. True
4. True
5. True

JOB SHEET 2.3-3
Replanting missing hills

Title: Replanting
Performance Objective: At the end of this module you should be able to replant missing hills.
Supplies/Materials: PPE, seedlings, trowel, garden plot
Steps/Procedures: wear the PPE. Identify missing hills Prepare seedling to be used in replanting Plant seed prepared seedlings on missing hills Do housekeeping.
Assessment method: **Demonstration:**

PERFORMANCE CHECKLIST 2.3-3
Replanting missing hills

Performance Criteria	YES	NO
Did the trainee .		
wear the PPE?		
Identify missing hills?		
Prepare seedling to be used in replanting?		
Plant seed prepared seedlings on missing hills?		
Do housekeeping?		

INFORMATION SHEET 2.3-4

Plant rationing

Learning Objective: After reading this information sheet, you should be able identify the number of sow to be served per boar.

Introduction:

Plant Rejuvenation- means restoring vitality and freshness of plants. It is another Name for renewal.

- Rejuvenation is attempted to make the plant new.
- In India many existing orchards are not as productive as their potential.
- Selection of poor planting material, improper plantation and upkeep make orchards uneconomic. These situations necessitate need for rejuvenation.
- Various factors which make the plant susceptible to rejuvenations are as under:

Techniques of Rejuvenation

Pruning

- Pruning -is very powerful technique of rejuvenating the plants. It brings juvenility. While pruning every attempt is made to remove dead, damaged, diseased and inter lacerating branches.
- Generally pruning very is resorted at breast height at about 2m height from ground level. Lower the pruning more vigorous is the sprouting. It is the physiological age not chronological age that is most important for rejuvenility and in turn the ability of the plant to sprout.
- Futher after long bearing age, trees itself enter into senility. Pruning helps in invigorating such trees.

SELF CHECK 2.3-4

Plant rationing

TRUE OR FALSE

1. _______ Plant Rejuvenation means restoring vitality and freshness of plants.
2. ________Pruning is very powerful technique of rejuvenating the plants.
3. ________Rejuvenation is attempted to make the plant new.
4. ________Lower the pruning more vigorous is the sprouting.
5. ________While pruning every attempt is made to remove dead, damaged, diseased and inter lacerating branches.

ANSWER KEY 2.3-4

Plant rationing

1. True
2. True
3. True
4. True
5. True

JOB SHEET 2.3-4

Plant rationing

Title: Rejuvenating/ rationing plants
Performance Objective: At the end of this module you should be able to perform the techniques on rationing plants.
Supplies/Materials: PPE, Pruning shear, container, crops
Steps/Procedures: Wear the PPE. Gather the tools and materials. Go to your garden. Check your crops if it's infested by pest and diseases Do pruning Do housekeeping.
Assessment method: **Demonstration:**

PERFORMANCE CHECKLIST 2.3-4

Plant rationing

Performance Criteria	YES	NO
Did the trainee ..		
Wear the PPE?		
Gather tools and materials?		
Going to your garden?		
Check your crops if it's infested by pest and diseases?		
Do pruning?		
Do housekeeping?		

INFORMATION SHEET 2.3-5

Organic fertilizers application

Learning Objective: After reading this information sheet, you should be able identify organic fertilizers.

Introduction:

Organic fertilizers are an essential source for plant nutrients and a soil conditioner in agriculture. Due to its sources and the composition of the organic inputs as well as the type, functionality and failures of the applied treatment process, the organic fertilizer may contain various amounts of infectious agents and toxic chemicals, especially the antibiotics that can be introduced to the subsequent food chain. A range of human and animal pathogens of bacterial, viral and parasitic origin have been the cause of food-borne epidemics due to unintended contamination from organic fertilizers. The use of antibiotics by humans and in animal feeds will also end up in the organic fertilizers. These antibiotics and other chemicals, depending on the sources of the organics, will enhance the likelihood of occurrence of resistant and multi-resistant strains of microorganisms in society and have been reported to cause ecotoxicological environmental effects and disruption of the ecological balance. Exposure of microorganisms to sublethal concentration of antibiotics in the organic products induces antibiotic resistance. WHO guidelines for the reuse of excreta and other organic matters identify the risk for the exposed groups to the reuse of the excreta and are applicable in the use of organic fertilizers in agriculture.

The potential health intricacies linked with organic fertilizers relate to their origin, their treatment and human exposure within a system perspective from origin to use, including products like crop type. Since organic fertilizers mainly are "faecal material/manure and urine from different animals and/or humans, with the addition of plant materials (organic solid wastes), or in special situations waste materials [1] from food or plant processing industries", the origin of the different fractions and their amounts partly defines the risk. Usually the risk is outbalanced by a wide range of benefits that the use of organic fertilizer exerts in agriculture as nutritional fertilizers and for soil conditioning. It has been further implied as more environmental friendly than the inorganic fertilizers [1] and its effect more tender on biotic components of the ecosystem without much shift in the ecological balance [2]. This is partly reflected by organisms like earthworms which may be negatively affected by inorganic fertilizers but promoted by the use of organic fertilizers and also incorporated as decomposers in aerobic composting processes [3, 4].

As this chapter deals with the public health aspects and risks involved, we define the organic materials utilized by its sources and thus relate to the following:

- Human faecal materials (also sludge from domestic treatment plants and from on-site sanitation, e.g. pit latrine emptying).
- Human urine (if separated).
- Animal manure (some risk differences depending on the species of animals/birds).
- Animal urine (often collected/spread separately, but impacted by the animal faeces).
- Other types of organic solid wastes (plant materials, domestic, industrial from organic food/fodder processing industries).

Additionally, the risk may relate to some storage-specific factors like

- Regrowth of specific bacterial pathogens or opportunistic ones (occurs when the material that, for example should be/are composted, are not well stabilized or broken down. During these circumstances, for example *Escherichia coli, Salmonella* sp., *Listeria* sp. and spore formers will regrow in the material if present).
- When the collected/stored/kept organic fractions or mixture thereof (see above) function as a breeding site for flies and mosquitoes that serve as vectors of parasitic diseases.
- Development of spore-forming thermophilic fungi and Actinomycetes in composting processes, where the spores can cause diseases in both immune-competent and immune-compromised individuals upon inhalation. An example of such an organism is *Aspergillus fumigatus*.

Based on source, the risk will vary to a great extent, depending on the health of the animals/humans that primarily defines the microbial concentration and partly occurrence of antibiotics and chemical components in the organic wastes (from domestic or animal sludge fractions) that may be conveyed to the agricultural sites and crops fertilized. Additional components may apply if organic industrial wastes are utilized. An indirect organic fertilization may occur through irrigation using wastewater effluent, where the nutrient load serves as an advantage. This is widely applied in developing countries [5]. However, this may result in additional inputs of antibiotics, toxic organic and inorganic compounds and pathogens. All these concepts are further deliberated in this chapter. The possibilities of recycling food-borne pathogens via agricultural crops to the final end consumers of the crops will additionally be discussed. Food-borne pathogens are especially important for animal faecal-based fertilizers used on fruits and vegetables farms meant to supply salads in restaurants. Other dynamics are residual antibiotics which are sometimes locked in the components of the organic fertilizers with attending public health implications to be further enumerated in this chapter.

SELF CHECK 2.3-5

Organic fertilizers application

TRUE OR FALSE

1. _________ Organic fertilizers are an essential source for plant nutrients and a soil conditioner in agriculture.
2. _________ The use of antibiotics by humans and in animal feeds will also end up in the organic fertilizers.
3. _________ Exposure of microorganisms to sub lethal concentration of antibiotics in the organic products induces antibiotic resistance.
4. _________ The possibilities of recycling food-borne pathogens via agricultural crops to the final end consumers of the crops will additionally be discussed.
5. __________ The potential health intricacies linked with organic fertilizers relate to their origin, their treatment and human exposure within a system perspective from origin to use, including products like crop type.

ANSWER KEY 2.3-5

Organic fertilizers application

1. TRUE
2. TRUE
3. TRUE
4. TRUE
5. TRUE

Learning Outcome #4 Perform harvest and Post-harvest activities

Assessment Criteria:

Products are checked using maturity indices according to PNS, PNS-organic agriculture and enterprise practice.

Marketable products are harvested according to PNS, PNSorganic agriculture and enterprise practice.

Harvested vegetables are classified according to PNS, PNSorganic agriculture and enterprise practice.

Appropriate harvesting tools and materials are used according to PNS.

Post-harvest practices are applied according to PNS and GAP recommendation

Production record is accomplished according to enterprise procedures

Contents

maturity indices

harvest marketable products

classify marketable products

harvesting tools and materials

Post-harvest practices

Record Keeping

Tools, Materials and Equipment and Facilities

Ladder
Basket
Scissors
Scythe
Notebook and pen

References:

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LEARNING EXPERIENCES/ACTIVITIES

LO # 4 Perform harvest and post-harvest activities

Learning Activities	Special Instructions
Read the attached Information Sheet 2.4-1 on Maturity indices	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 2.4-1 on Maturity indices	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 2.4-1 on Maturity indices	
Read the attached Information Sheet 2.4-2 on Harvest marketable products	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 2.4-2 on Harvest marketable products	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 2.4-2 on Harvest marketable products	
Read the attached Information Sheet 2.4-3 on Classify marketable products products	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 2.4-3 on Classify marketable products	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 2.4-3 on Classify marketable products	
Read the attached Information Sheet 2.4-4 on Harvesting tools and materials	You can ask the assistance of your trainer explain further the topics you cannot understand

Answer Self-Check 2.4-4 on Harvesting tools and materials	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 2.4-4 on Harvesting tools and materials	
Read the attached Information Sheet 2.4-5 on Post-harvest practices	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 2.4-5 on Post-harvest practices	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 2.4-5 on Post-harvest practices	
Read the attached Information Sheet 2.4-6 on Record Keeping	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 2.4-6 on Record Keeping	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 2.4-6 on Record Keeping	

INFORMATION SHEET 2.4-1
Maturity Indices

Learning Objective: After reading this information sheet, you should be able identify the maturity indices of fruits of vegetables.

Introduction:

Maturity Indices – are the sign or indication the readiness of the commodity for harvesting

The principles dictating at which stage of maturity a fruit or vegetable should be harvested are crucial to its subsequent storage and marketable life and quality. Post-harvest physiologists distinguish three stages in the life span of fruits and vegetables: maturation, ripening, and senescence. Maturation is indicative of the fruit being ready for harvest. At this point, the edible part of the fruit or vegetable is fully developed in size, although it may not be ready for immediate consumption. Ripening follows or overlaps maturation, rendering the produce edible, as indicated by taste. Senescence is the last stage, characterized by natural degradation of the fruit or vegetable, as in loss of texture, flavour, etc. (senescence ends at the death of the tissue of the fruit). Some typical maturity indexes are described in following sections.

Skin colour:

This factor is commonly applied to fruits, since skin color changes as fruit ripens or matures. Some fruits exhibit no perceptible colour change during maturation, depending on the type of fruit or vegetable. Assessment of harvest maturity by skin colour depends on the judgment of the harvester, but colour charts are available for cultivars, such as apples, tomatoes, peaches, chilli peppers, etc.

Optical methods:

Light transmission properties can be used to measure the degree of maturity of fruits. These methods are based on the chlorophyll content of the fruit, which is reduced during maturation. The fruit is exposed to a bright light, which is then switched off so that the fruit is in total darkness. Next, a sensor measures the amount of light emitted from the fruit, which is proportional to its chlorophyll content and thus its maturity.

Shape:

The shape of fruit can change during maturation and can be used as a characteristic to determine harvest maturity. For instance, a banana becomes more rounded in cross-sections and less angular as it develops on the plant. Mangoes also change shape during maturation. As the mango matures on the tree the relationship between the shoulders of the fruit and the point at which the stalk is attached may change. The shoulders of immature mangoes slope away from the fruit stalk; however, on more mature mangoes the shoulders become level with the point of attachment, and with even more maturity the shoulders may be raised above this point.

Size:

Changes in the size of a crop while growing are frequently used to determine the time of harvest. For example, partially mature cobs of *Zea mays saccharata* are marketed as sweet corn, while even less mature and thus smaller cobs are marketed as baby corn. For bananas, the width of individual fingers can be used to determine harvest maturity. Usually a finger is placed midway along the bunch and its maximum width is measured with callipers; this is referred to as the calliper grade.

Aroma:

Most fruits synthesize volatile chemicals as they ripen. Such chemicals give fruit its characteristic odor and can be used to determine whether it is ripe or not. These doors may only be detectable by humans when a fruit is completely ripe, and therefore has limited use in commercial situations.

Fruit opening:

Some fruits may develop toxic compounds during ripening, such as ackee tree fruit, which contains toxic levels of hypoglycine. The fruit splits when it is fully mature, revealing black seeds on yellow arils. At this stage, it has been shown to contain minimal amounts of hypoglycine or none at all. This creates a problem in marketing; because the fruit is so mature, it will have a very short post-harvest life. Analysis of hypoglycine 'A' (hyp.) in ackee tree fruit revealed that the seed contained appreciable hyp. at all stages of maturity, at approximately 1000 ppm, while levels in the membrane mirrored those in the arils. This analysis supports earlier observations that unopened or partially opened ackee fruit should not be consumed, whereas fruit that opens naturally to over 15 mm of lobe separation poses little health hazard, provided the seed and membrane portions are removed. These observations agree with those of Brown et al. (1992) who stated that bright red, full sized ackee should never be forced open for human consumption.

Leaf changes:

Leaf quality often determines when fruits and vegetables should be harvested. In root crops, the condition of the leaves can likewise indicate the condition of the crop below ground. For example, if potatoes are to be stored, then the optimum harvest time is soon after the leaves and stems have died. If harvested earlier, the skins will be less resistant to harvesting and handling damage and more prone to storage diseases.

Abscission:

As part of the natural development of a fruit an abscission layer is formed in the pedicel. For example, in cantaloupe melons, harvesting before the abscission layer is fully developed results in inferior flavoured fruit, compared to those left on the vine for the full period.

Firmness:

A fruit may change in texture during maturation, especially during ripening when it may become rapidly softer. Excessive loss of moisture may also affect the texture of crops. These textural changes are detected by touch, and the harvester may simply be able to gently squeeze the fruit and judge whether the crop can be harvested. Today sophisticated devices have been developed to measure texture in fruits and vegetables, for example, texture analyzers and pressure testers; they are currently available for fruits and vegetables in various forms. A force is applied to the surface of the fruit, allowing the probe of the penetrometer or texturometer to penetrate the fruit flesh, which then gives a reading on firmness. Hand held pressure testers could give variable results because the basis on which they are used to measure firmness is affected by the angle at which the force is applied. Two commonly used pressure testers to measure the firmness of fruits and vegetables are the Magness-Taylor and UC Fruit Firmness testers (Figure 2.1). A more elaborate test, but not necessarily more effective, uses instruments like the Instron Universal Testing Machine. It is necessary to specify the instrument and all settings used when reporting test pressure values or attempting to set standards.

SELF CHECK 2.4-1

Maturity Indices

TRUE OR FALSE:

1. Skin color changes as fruit ripens or mature.
2. Changes in the size of a crop while growing are frequently used to determine the time of harvest.
3. The shape of fruit can change during maturation and can be used as a characteristic to determine harvest maturity.
4. Most fruits synthesize volatile chemicals as they ripen. Such chemicals give fruit its characteristic odor and can be used to determine whether it is ripe or not.
5. Maturity Indices – are the sign or indication the readiness of the commodity for harvesting

ANSWER KEY 2.4-1

Maturity Indices

1. TRUE
2. TRUE
3. TRUE
4. TRUE
5. TRUE

INFORMATION SHEET 2.4-2

Harvest Marketable products

Learning Objective: After reading this information sheet, you should be able to harvest marketable products.

Introduction:

When to Harvest/Maturity Guidelines:
• Harvest before the plant flowers or sends up a flower spike.
• Leaves should be tender, not tough.
• For arugula and salad mix, cut when leaves are small and very tender (3-4 inches high); be sure weeds or grasses have been removed from harvest area.
• For pea tendrils, snap off tips of plants where stem is still succulent and soft; harvest before stems turn stiff or woody. Time of Day to Harvest: • Early morning is the best time to harvest most greens, since that is the coolest time of day. • If morning is not possible, try to harvest in the evening after the heat of the day has passed.

Fruit and Melons

Asian Cucumbers Bitter Melon Cucumbers Eggplant, all varieties Kittely Peppers, all varieties Summer Squash Tomatillos Tomatoes Zucchini
When to Harvest/Maturity Guidelines:
• Harvest when fruit is the desirable size and/or color, and when the flesh is firm but ripe.
• For cucumbers, eggplant, melon, and squash, do not harvest too late, or fruit can become bitter and/or seedy. Time of Day to Harvest:
• Early morning is the best time to harvest, while it is still cool but after dew has dried from the fruit. 7:00am-9:00am Harvesting and Post-Harvest Handling Instructions:
• Cucumbers, melons, and tomatoes can be picked by turning the fruit parallel to the stem and quickly snipping it off.
• Use scissors or clippers to cut the stems of eggplant, peppers, and squash just above the fruit.
• Keep fingernails trimmed when harvesting squash to avoid punctures.

• These crops are delicate - be careful not to puncture or bruise during harvest.
• Do not stack too many fruits in one bin or those at the bottom will be crushed.
• Keep vegetables in the shade and cool as soon as possible.
• No need to rinse or wash unless dirt has adhered to the vegetables.

Cruciferous Vegetables

Baby Bok Choy Broccoli Broccoli Raab Chinese Broccoli Cabbage Pac Choy
When to Harvest/Maturity Guidelines:
• Harvest baby bok choy or pac choy when the leaves are about 6-10 inches heigh. Cut just above the root so the head stays together.
• Broccoli should be harvested when the heads are fully formed but still tight and compact - before flowering. If you leave the plant in the ground more baby broccoli heads will grow off the main stalk that you can keep harvesting.
• Broccoli raab and Chinese broccoli should be harvested when the leaves are still tender and flower heads are formed. Chinese broccoli may be starting to flower at time of harvest.
• Pick cabbage when the heads are fully formed and firm. Each head should weigh at least one pound. Time of Day to Harvest:
• Cooler times of day, if possible (mornings & evenings)
Harvesting and Post-Harvest Handling Instructions:
• Baby bok choy and pac choy should be cut just above the roots at ground level. Trim off any yellow leaves. Bunch 2-4 heads together and fasten with a rubber band.
• Cut broccoli raab and Chinese broccoli stems about 8 inches long.
• Cut broccoli about 4-6 inches below the head. Do not leave a very long stem. Remove large leaves from the stem.
• Cut cabbages closely below the head. Remove large outer leaves, but do not peel off too many leaves or the cabbage will spoil more quickly.

Roots and Tubers

Beets Carrots Potatoes Sweet Potatoes Radishes Turnips
When to Harvest/Maturity Guidelines:
• Harvest carrots based on the particular variety’s size guidelines (read the seed packet), usually when they are 5-10 inches long and about 1 inch wide. Lift the soil to loosen the carrots to pull them up with greens attached.
• Potatoes are usually harvested in the fall when the tops die and turn brown. Some potatoes, however, are harvested when they are small, about the size of a golf ball, and before the plant flowers. These are called ‘new’ potatoes.
• Sweet potatoes should be harvested in the fall before the first freeze.

• Radishes can be harvested when they are between 3/4 and 1 1/4 inches across. The radish should be crisp and the skin should not be cracked or split. Do not allow to grow too large or it may become hollow or "pithy" inside.

• Storage Turnips should be harvested when they are more than 1inch across. For young turnips (salad turnips or Hakurei variety), harvest beginning when turnips are the size of a small radish up to 1 inch across. Leave the stems and edible leaves attached to the turnip. Harvesting and Post-Harvest Handling Instructions:

• Be careful not to slice root crops if you use a shovel or fork to loosen the soil.

• Carrots and radishes should be sprayed with a hose or rinsed to remove all dirt then tied into bunches with twist ties

. • Turnips and beets can be sold with or without their edible tops. If you are leaving the tops on, remove yellow or damaged leaves. Rinse or spray with a hose to remove dirt, then tie into bunches with twist ties. You may want to sell the bunched top greens separately

• Potatoes should be harvested and left in a cool, dry place for several days to cure. After they are cured, you can brush or wash off the dirt. Potatoes store best when not pre-washed.

• Sweet potatoes should be harvested and left in a warm, humid place for a few days to cure. After they are cured, you can brush or wash off the dirt.

SELF CHECK 2.4-2

Harvest Marketable products

TRUE OR FALSE

1. Harvest when fruit is the desirable size and/or color, and when the flesh is firm but ripe.
2. Early morning is the best time to harvest most greens, since that is the coolest time of day.
3. Use scissors or clippers to cut the stems of eggplant, peppers, and squash just above the fruit.
4. Pick cabbage when the heads are fully formed and firm. Each head should weigh at least one pound. Time of Day to Harvest.
5. Keep fingernails trimmed when harvesting squash to avoid punctures.

ANSWER KEY 2.4-2

Harvest Marketable products

1. TRUE
2. TRUE
3. TRUE
4. TRUE
5. TRUE

JOB SHEET 2.4-2

Harvest marketable products

Title: Performing harvesting vegetables
Performance Objective: At the end of this module you should be able to perform proper harvesting in vegetables
Supplies/Materials: PPE, harvesting tools, containers
Steps/Procedures: Wear the PPE. Gather the tools and materials Going to the field Check the appropriate maturity index of your crops Perform Harvesting Do housekeeping.
Assessment method: **Demonstration:**

PERFORMANCE CHECKLIST 2.4-
Harvest marketable products

Performance Criteria	YES	NO
Did the trainee ..		
Wear the PPE?		
Gather the tools and materials?		
Going to the field?		
Check the appropriate maturity index of your crops?		
Perform Harvesting?		
Do housekeeping?		

INFORMATION SHEET 2.4-3

Classify marketable products

Learning Objective: After reading this information sheet, you should be able to classify marketable products.

Introduction:

Practices that are critical to managing produce safety and quality during production, harvesting and postharvest handling are identified for the crop grown.

Appropriate maturity indices should be the bases in determining the harvest time. Appropriate harvesting technique should be employed in harvesting to optimize the quality and other desired characteristics of produce during harvest or postharvest phases.

Harvesting time should be done in accordance to commodity requirements. Harvesting under the rain should be avoided. Fresh fruits and vegetables that are unfit for human consumption should be segregated during harvesting. Those which cannot be made safe by further processing should be disposed properly to avoid contamination of the uncontaminated produce.

Containers used for harvesting should be suitable and clean before use. Liners are preferably used to protect the produce, particularly when containers have rough surfaces. 4.7.6 If the containers are recycled, these should be properly cleaned or discarded accordingly if found unfit for use.

Harvested produce should not be placed in direct contact with the soil or floor in the handling, packing or storage areas. Packaging 4.7.8 Produce should be graded and packed according to market requirements. 4.7.9 When packing of fresh fruits and vegetables is done in the field, contaminated containers or bins exposed to the sources of contaminants (i.e. manure) should be avoided. 4.7.10 Protective materials should be used whenever appropriate to protect the produce from rough surfaces of containers and exposure to sunlight leading to excessive moisture loss.

SELF CHECK 2.4-3

Classify marketable products

True or False

1. _________________ Containers used for harvesting should be suitable and clean before use
2. _________________ Appropriate maturity indices should be the bases in determining the harvest time.
3. _________________ Appropriate harvesting technique should be employed in harvesting to optimize the quality and other desired characteristics of produce during harvest or postharvest phases.
4. __________________ Fresh fruits and vegetables that are unfit for human consumption should be segregated during harvesting.
5. _________________ Harvested produce should not be placed in direct contact with the soil or floor in the handling, packing or storage areas.

ANSWER KEY 2.4-3
Classify marketable products

1. True
2. True
3. True
4. True
5. True

INFORMATION SHEET 2.4-4

Harvesting tools and materials

Learning Objective: After reading this information sheet, you should be able identify the best tools for harvesting.

Introduction:

Having the right set of tools sometimes makes the difference between an enjoyable interlude in the garden or orchard and sweaty hours of backbreaking labor. The proper tools make bringing in the harvest easier, faster and simply more fun. Here are some items to consider to help you get the job done.

Garden Scissors

If you, like me, harvest your own herbs, garden scissors could be your new best friend. Also known as flower shears, most garden scissors feature narrow, sharply pointed anvil blades that make snipping between tight stems a breeze. Things to look for include comfortable handles (soft plastics are especially easy on the hands) and durable, stay-sharp stainless-steel blades.

Joyce Chen Unlimited Garden Scissors are typical of this breed. They feature soft-grip handles; tapered points for precision cutting; and finely honed blades of chrome molybdenum stainless steel. Their design makes them ideal for either right- or left-handed use and many retail outlets stock them.

Fiskars markets a slightly different, but-just-as-efficient, design. Fiskars Garden Shears have extra-sharp, serrated, hardened stainless steel blades that grip and hold plant material for clean cuts; comfortable, ergo-dynamic handles; and come with a lifetime warranty!

Digging Tools

You'll never appreciate a precision digging tool more than you do at harvest time—especially the classic Spear and Jackson County Trowel for hand-digging bulbs and roots. Skillfully crafted, perfectly balanced and ruggedly dependable, this heirloom-quality trowel has a carbon-steel head and a weatherproofed, hardwood handle for greater durability.

I once dug a large garden plot of potatoes with a pitchfork—never again! The right tool for the right job, I say after that ordeal—and that tool is a proper digging fork. Spear and Jackson make especially fine ones; their Neverbend Professional Potato Fork has extra-wide, forged carbon-steel tines specifically designed for lifting potatoes. Its entire head is epoxy-coated for improved resistance to rust and scratches; a traditional hardwood shaft completes this pretty picture.

Garden Carry-all

You can carry goodies in from the garden and orchard in less picturesque containers, but why? Patterned after traditional New England clam baskets, Pike's Original Maine Garden Hods, manufactured by Maine Garden Products, are a cut way above a bucket or sack. Pike's carry-alls are crafted of pine and steam-bent oak, with birch side rods and a food-grade, PVC-coated wire mesh body that makes it easy to rinse your crops right in the hod. They come in two handy sizes, the 16-quart Original Hod and the 8-quart L'il Hod in logo-branded, plain and painted models.

Scythe

If you'd like to hand-harvest hay for your goats, horses or rabbits, here is your tool. According to Scythe Supply—one of America's leading authorities on European-style scythes—you can easily scythe enough grass over the summer to put up hay for eight to 10 goats. The trick is buying a quality scythe and learning to maintain and use it. They say the scythe isn't that difficult to master and their informative website (www.scythesupply.com) is loaded with resources to teach you how to do it.

Nut and Dropped-fruit Harvester

If you've ever wanted to harvest dropped fruit or nuts that litter your yard each fall, but dread hours of bending over to pick them up, the Nut Wizard is the tool for you. The Nut Wizard, originally invented for harvesting pecans and walnuts, is a revolving, spring-wire cage on a handle. Capable of picking up most any object between 3/8-inch and 4 inches in diameter, the 41⁄2-foot-long, 3- to 5-pound tool (depending on the size) requires very little pressure to operate and comes in three sizes, the better to handle the precise sorts of dropped nuts and fruits in your own yard.

Fruit Pickers

If you need to pick fruit from trees instead of off the ground, there are multiple tools designed to help do the trick.

An innovative design is the Twister Picker, a lightweight (8 ounces), aluminum-and-plastic picker tool designed to be mounted on a pole, such as a standard broom handle. To use it, the operator clamps the aluminum fruit holders on a single piece of fruit and twists it off by rotating the pole in her hands. Fruit is picked undamaged with this device and it almost makes fruit picking fun!

Ames' lightweight, basket-style Fruit Picker's bent-wire fingers are designed to coax apples, oranges, plums, peaches, avocadoes and more from trees. Its picker head comes fitted with a two-piece, wooden handle; Ames applies an enamel finish to protect the picker's wire basket from rust and corrosion, and adds a foam pad inside the basket to minimize bruising of valuable fruit.

The German company, Wolf-Garten, is known for its quality tools, among them the Wolf-Garten RG-M Fruit Picker. Its adjustable picker head's strong, nylon fingers grip each fruit without bruising it; then a concealed cutting blade near the front of the basket cuts it off the tree; finally it tumbles into the unit's soft, four-apple capacity canvas bag. The picker head is used with the buyer's choice of Wolf-Garten interchangeable Interlocken System expandable handles, sold separately; Wolf-Garten has over 50 different attachments that work with the Interlocken System.

Orchard Ladders

If you have so much fruit that you need a ladder to pick it, don't settle for the rickety, old ladder in your garage! Household ladders aren't designed for orchard use; play it safe—buy a tool designed for the job.

Orchard ladders come in several designs, including four-legged, double-step ladders, straight ladders designed to lean against a tree and the créme de la créme of orchard ladders: the three-legged, traditional tripod orchard ladder.

Which-ever you choose, opt for a sturdy, lightweight, tempered aluminum ladder with each step braced for maximum security and one tall enough to do the job you have in mind (you must never stand above the third rung from the top of an orchard ladder, so plan accordingly). Keep in mind that tripod ladders are designed for use on soil or grassy surfaces only, so don't choose this style to use for any type of indoor application.

Ladder King manufactures a full line of aluminum orchard ladders, including the Ladder King 1400 Series Double Step Ladders designed for two-person use in 3- to 10-foot heights. Three-inch-wide steps with raised ribbing to prevent slippage; heavy-duty bracing to minimize twisting; strong spreader hinges to hold the ladder steady during use; and sure-grip rubber feet are all features of the line.

Tallman, a manufacturer of orchard ladders, features a tripod orchard ladder safety video at www.tallmanladders.com Tallman's elegant tripod ladders come in 4- to 16-foot lengths with slip-resistant steps and rigid steel hinges for strength and durability.

Garden Carts

I've saved my most indispensable harvest tool for last: a sturdy, easy-to-push- (or pull) and-maneuver garden cart. I simply couldn't farm without one. There are scores of styles ideal for every need.

When choosing a garden cart, remember these points:

- Some carts push and maneuver infinitely easier than others; it's always best to try before you buy.
- The more you plan to use your cart, the sturdier it must be.
- If you'll need to maneuver the cart through a doorway or gate, measure to make certain it will fit.
- Combination wood and metal carts require indoor storage; metal or plastics usually weather well
- Carts with pneumatic tires generally push easiest, but flat-free, solid tires are better for jobs like harvesting walnuts in the woods.

SELF CHECK 2.4-4
Harvesting tools and materials

True or False

1. _________________ Having the right set of tools sometimes makes the difference between an enjoyable interlude in the garden or orchard and sweaty hours of backbreaking labor.

2_________________ The proper tools make bringing in the harvest easier, faster and simply more fun.

3.__________________ Carts with pneumatic tires generally push easiest.

4. __________________ Flat-free carts solid tires are better for jobs like harvesting walnuts in the woods.

5. _________________ Combination wood and metal carts require indoor storage; metal or plastics usually weather well

ANSWER KEY 2.4-4
Harvesting tools and materials

1. True
2. True
3. True
4. True
5. True

INFORMATION SHEET 2.4-5

Post-harvest practices

Introduction:

Postharvest handling is the stage of crop production immediately following harvest, including cooling, cleaning, sorting and packing. The instant a crop is removed from the ground, or separated from its parent plant, it begins to deteriorate. Postharvest treatment largely determines final quality, whether a crop is sold for fresh consumption, or used as an ingredient in a processed food product.

The most important goals of post-harvest handling are keeping the product cool, to avoid moisture loss and slow down undesirable chemical changes, and avoiding physical damage such as bruising, to delay spoilage.[1] Sanitation is also an important factor, to reduce the possibility of pathogens that could be carried by fresh produce, for example, as residue from contaminated washing water.

After the field, post-harvest processing is usually continued in a packing house. This can be a simple shed, providing shade and running water, or a large-scale, sophisticated, mechanised facility, with conveyor belts, automated sorting and packing stations, walk-in coolers and the like. In mechanised harvesting, processing may also begin as part of the actual harvest process, with initial cleaning and sorting performed by the harvesting machinery.

Initial post-harvest storage conditions are critical to maintaining quality. Each crop has an optimum range of storage temperature and humidity. Also, certain crops cannot be effectively stored together, as unwanted chemical interactions can result. Various methods of high-speed cooling, and sophisticated refrigerated and atmosphere-controlled environments, are employed to prolong freshness, particularly in large-scale operations.

Postharvest shelf life

Once harvested, vegetables and fruits are subject to the active process of degradation. Numerous biochemical processes continuously change the original composition of the crop until it becomes unmarketable. The period during which consumption is considered acceptable is defined as the time of "postharvest shelf life".

Postharvest shelf life is typically determined by objective methods that determine the overall appearance, taste, flavor, and texture of the commodity. These methods usually include a combination of sensorial, biochemical, mechanical, and colorimetric (optical) measurements. A recent study attempted (and failed) to discover a biochemical marker and fingerprint methods as indices for freshness

Post-harvest physiology

Post-harvest physiology is the scientific study of the physiology of living plant tissues after picking. It has direct applications to postharvest handling in establishing the storage and transport conditions that best prolong shelf life.

An example of the importance of the field to post-harvest handling is the discovery that ripening of fruit can be delayed, and thus their storage prolonged, by preventing fruit tissue respiration. This insight allowed scientists to bring to bear their knowledge of the fundamental principles and mechanisms of respiration, leading to post-harvest storage techniques such as cold storage, gaseous storage, and waxy skin coatings

STEPS IN POST HARVEST PRACTICES

Trimming

Washing fruits and vegetables fresh from the field in a properly equipped dunk tank or hydrocooler can be of tremendous benefit, both in extending the shelf life of the produce and improving its safety to the consumer. If not properly managed, washing produce can amplify a small problem into a big one. There are a few points to consider for proper washing of fresh produce.

Sorting is done by hand to remove the fruits and vegetables which are unsuitable to market or storage due to damage by mechanical injuries, insects, diseases, immature, over-mature, misshapen etc. This is usually carried out manually and done before washing. By removing damaged produce from the healthy ones, it reduces losses by preventing secondary contamination. Sorting is done either at farm level or in the pack-houses. In sorting, only sensory quality parameters are taken into consideration.

Weighing

Packaging means the wrapping or bottling of products to make them safe from damages during transportation and storage. It keeps a product safe and marketable and helps in identifying, describing, and promoting the product.

Labeling

Delivery

SELF CHECK 2.5-5

Post-harvest practices

True or False

1. Postharvest handling is the stage of crop production immediately following harvest, including cooling, cleaning, sorting and packing.
2. The most important goals of post-harvest handling are keeping the product cool, to avoid moisture loss and slow down undesirable chemical changes, and avoiding physical damage such as bruising, to delay spoilage.
3. Post-harvest physiology is the scientific study of the physiology of living plant tissues after picking.
4. **Packaging** means the wrapping or bottling of products to make them safe from damages during transportation and storage.
5. By removing damaged produce from the healthy ones, it reduces losses by preventing secondary contamination.

ANSWER KEY 2.5-5

Post-harvest Practices

1. TRUE
2. TRUE
3. TRUE
4. TRUE
5. TRUE

INFORMATION SHEET 2.4-6

Record Keeping

Introduction:

There are three sets of basic records that should be kept by the owner of a small fruit and vegetable processing unit: financial records, those that relate to the production of the products and sales records. The uses of these records are inter-related and are described in more detail in Sections 2.3 and 2.7. In this Section, the format of the records and the likely ways in which information will be obtained are summarized.

As with all other inputs to a business, keeping records is an investment of time and money and the benefits must outweigh the costs. There is no point in recording information for its own sake and records must be used if they are to have any value. This means that the owner or manager must understand why the information is collected and what it can be used for. Similarly, the time and effort spent in keeping records must be related to the scale and profitability of the business. While it is true that some successful entrepreneurs keep all of the information in their head and do not keep records, no-one else can help run the business during times of illness or absence. Some examples of the value and costs of keeping records are shown below:

Value of record keeping:

- detailed knowledge about the operation of the business
- identification of trends
- accurate control over finances and product quality
- identification of individual costs to allow changes to a product or process to optimise profits
- keeping track of money owed to the business
- evidence for tax authorities (may be a legal requirement)
- factual basis for product pricing or salary levels
- knowledge and avoidance of theft.

Costs of record keeping:

 time spent learning how to keep records or training staff time spent writing them

 cost of materials such as ledgers and pens

 information is written down and therefore potentially available for competitors or authorities to see

 cost of keeping records private and secure.

Accurate information is essential and this means that staff who are required to collect information should know its value and why it is being collected. This should be part of the induction and training when new staff learn their job. The entrepreneur should employ people who have the skills and aptitude to do the work, but should also put in place a system of checks to ensure that one person does not have responsibility for a whole area of business activity. For example the person responsible for keeping records of purchases should be different from the person who records use of materials or levels of stocks. The owner or manager should also ensure that all records are kept up to date and where appropriate, the arithmetic is checked for accuracy. There is no single correct way to keep records and individual owners should devise systems that suit their way of working. The examples given below have been found to be successful in small food processing enterprises in Africa and Asia.

2.9.1. Financial and sales records

A separate record of the cash that comes into a business and the cash that is used to buy daily items is usefully prepared using a *Cash Book* (Figure 62). Additionally, when entrepreneurs have a bank account, they will require a *Bank Book* to record cheques that have been received and paid, using the same headings as those shown in Figure 62.

Figure 62. - Example of a Cash Book layout

Date	Item	Cash in	Cash out	Balance
____	*Description or invoice number*	______	________	_______
____	________________________	______	________	_______
____	________________________	______	________	_______

It is important to know how much money the business is owed by debtors at any given time but also how much is owed to creditors. This is particularly important if for example, retailers expect a period of credit before they pay for goods received. The amount of money owed by an enterprise and the amount owing to it can be combined in a single ledger so that a weekly comparison of the difference can be made. Invoices and receipts should be kept together in date order. An example of this type of ledger is an *Accounts Receivable and Payable Book* (Figure 63).

Figure 63. - Example of an Accounts Receivable and Payable Book

Date	Item	Credit given	Balance	Date	Item	Credit taken	Balance
____	*Description or receipt number*	______	*Description or invoice number*	_____	_____	___________	______
____	______________	______	____________	_____	_____	___________	______
____	______________	______	____________	_____	_____	___________	______
____	______________	______	____________	_____	_____	___________	______

Other books can be used but these are the basic requirement for collecting all financial information needed to prepare monthly profit and loss statements, balance sheets and to check cashflow forecasts. The other information needed to prepare profit and loss statements are records of sales and stock in the storerooms (Figures 64 and 65).

Figure 64. - Example of a page from a Sales Book

Product Name ________________________________ **Batch Number** _____________

Date	Customer	Amount Sold	Value	Invoice Date	Payment Date
____________	*Write in amount in kg or number of packs*	___________	________	______	_______
____________	_________________	___________	________	______	_______
____________	_________________	___________	________	______	_______
____________	_________________	___________	________	______	_______

Records that are kept by storekeepers show which products and materials are transferred into and out of the store-rooms. The balance is used to indicate when reordering is needed and can also be used to highlight pilferage or other losses that are not accounted for.

Figure 65. - Example of a Storekeeper's Book to keep account of ingredients (similar entries are made for packaging, materials and finished products)

Ingredient name

Date	Amount to store	Amount from store	Process batch number	Balance
______	*Write amounts in kg or number of packs*		______	______
______	______	______	______	______
______	______	______	______	______
______	______	______	______	______

As shown in Figure 66, data from the sales book is totalled to give monthly income. The costs of ingredients, packaging etc., that were used during the month are recorded in the storekeeper's book and other expenses are totalled from the cash book and bank book to calculate the monthly Profit and Loss Account.

The Profit and Loss Account describes how money comes into and leaves a business over a month (or other suitable period of time). This allows the owner to plot the progress of the business and compare the results to those expected in the Business Plan (Section 2.3.4).

However, to obtain a 'snapshot' of the performance of the business at a given moment, a *balance sheet* is a strong management tool which can help to understand where money came from, how it is used in a business and how it could be better used. An example of a balance sheet from a small wine-making business is shown in Figure 67.

Figure 66. - Example of a monthly Profit and Loss Account

Month: April

Item	**$**	**$**
Income from sales		750 *from Sales Book*
Less costs		
Opening stocks	25	
Purchases	55	
Stocks at end of month	30	
= Stocks used during month	50	*from Storeroom Ledger*
Gross profit		700
Less other expenses:		
Salaries	75	

Rent	25	
Supplies	20	*from cash and bank books*
Transport	45	
Marketing costs	25	
Interest repaid to lender	18	
Taxes paid	6	
	214	
Net profit	486	

The balance sheet is therefore a statement about the money in a business at a particular time, which shows how the money is being used (the assets) and where it came from (the liabilities). In the above example, the money that remains in the business as unclaimed profits is a main source of working capital. It is important to note that the owner has already taken a salary from the business and that the remaining profit belongs to the business to be used for reinvestment. This picture of the business can be used to determine, for example, whether more stock should be ordered, whether unpaid invoices to retailers should be followed up urgently or whether there are sufficient profits to repay a larger amount from the loan.

Figure 67. - Example of a balance sheet for a small wine-maker on 21.3.1997

Where money came from (Liabilities) ($)		**How the money was used (Assets) ($)**	
Accounts payable	450	Cash	65
Customer payments	865	Accounts receivable	650
Bank loan	1200	Stocks	600
Owner's capital	500	Equipment	2180
Profits	880	Owner's salary	400
	3895		3895

2.9.2. Production records

The main reasons for production records are to ensure that quality assurance procedures are in place and operating satisfactorily and to record the use of ingredients and amounts of stock for use in financial accounting. When raw materials are processed, each batch should be recorded in an *Incoming Materials Test Book* (Figure 68). The same layout can be used for recording incoming batches of ingredients and packaging materials, some of which also require inspection on arrival (see Section 2.7.2).

Records should also be kept of the amount and type of raw materials and ingredients that are used and the important processing conditions (e.g. drying times, heating times and temperatures etc.) to ensure that operators mix together the same ingredients in every batch and process them in the same way each time (Figure 69).

Each batch of food should be given a Batch Number which is recorded in stock control books, processing logbooks and product sales records. The batch numbers should be correlated with the product code numbers that are printed on labels or outer cartons. This allows the processor to trace any subsequent faults in a batch of product back to the process or to the raw materials.

Figure 68. - Example of an Incoming Materials Test Book

Product Name ____________________ **Batch Number** ____________

Raw material*	**Supplier**	**Results of inspection for**:**		
		A	**B**	**C**
Write in either 'Pass/Fail' or observations on quality				
____________	______	__________	____________	_________
____________	______	__________	____________	_________
____________	______	__________	____________	_________

* This information should also be recorded for other ingredients and packaging materials
** Different tests are written in place of ABC according to the types of materials being inspected

Figure 69. - Example of a Process Logbook for jam production

Product ___________________________ Batch Number __________________

Ingredients:	Target	Check	Changes from target
Pulp	*Write in amounts, times etc.*		_________________
Sugar	__________	__________	_________________
Citric Acid	__________	__________	_________________
Pectin	__________	__________	_________________
Batch weight	__________	__________	_________________
Boiling time	__________	__________	_________________
Solids content of product	__________	__________	_________________

In summary, therefore, record keeping is a management tool to help the owner to know the state of a small fruit and vegetable enterprise at any time and to have reliable information on which to base his or her plans for development of the business.

SELF CHECK 2.4-6
Record Keeping

TRUE OR FALSE

1. Financial records, those that relate to the production of the products and sales records.
2. Keeping records is an investment of time and money and the benefits must outweigh the costs.
3. Records that are kept by storekeepers show which products and materials are transferred into and out of the store-rooms.
4. The Profit and Loss Account describes how money comes into and leaves a business over a month (or other suitable period of time
5. The main reasons for production records are to ensure that quality assurance procedures are in place and operating satisfactorily and to record the use of ingredients and amounts of stock for use in financial accounting.

ANSWER KEY 2.4-6
Record Keeping

1. TRUE
2. TRUE
3. TRUE
4. TRUE
5. TRUE

ANSWER KEY 1.5-4

Record Keeping